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Real-time Optimal Braking for Marine Vessels with Rotating Thrusters

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Abstract

Collision avoidance is an essential component of autonomous shipping. As ships begin to advance towards autonomy, developing an advisory system is one of the first steps. An advisory system with a strong collision avoidance component can help the crew act more quickly and accurately in dangerous situations. One way to avoid collision is to make the vessel stop as fast as possible. In this work, two scenarios are studied, firstly, stopping along a predefined path, and secondly, stopping within a safe area defined by surrounding obstacles. The first scenario was further worked with to formulate a real-time solution.

Movements of a vessel, described in three degrees of freedom with continuous dynamics, were simulated using mathematical models of the forces acting on the ship. Nonlinear optimal control problems were formulated for each scenario and solved numerically using discretization and a direct multiple shooting method. The results for the first problem showed that the vessel could stop without much deviation from the path. Paths with different curvatures were tested, and it was shown that a slightly longer distance was traveled when the curvature of the path was greater. The results for the second problem showed that the vessel stays within the safe area and chooses a relatively straight path as the optimal way of stopping. This results in a shorter distance traveled compared to the solution of the first problem.

Two different real-time approaches were formulated, firstly a receding-horizon approach and secondly a lookup-based approach. Both approaches were solved with real-time feasibility, where the receding-horizon approach gave a better solution while the lookup-based approach had a shorter computational time.

Keywords

Nonlinear optimization, nonlinear optimal control, numerical optimal control, real-time solution, optimal braking, receding horizon, modeling, ship model, propeller forces, rudder forces, direct multiple shooting method, discretization.

Sammanfattning

Optimal bromsning i realtid för marina fartyg med roterande thrusters

Att undvika kollisioner är en viktig del av autonom sjöfart. När fartyg blir mer autonoma är utvecklingen av ett rådgivningssystem ett av de första stegen. Ett rådgivningssystem med en stark kollisionsundvikande komponent kan hjälpa besättningen att agera snabbare och mer exakt i farliga situationer. Ett sätt att undvika kollision är att få fartyget att stanna så snabbt som möjligt. I detta arbete studeras två scenarier, dels att stanna längs en fördefinierad bana samt att stanna inom ett säkert område avgränsat av omgivande hinder. Det första scenariot arbetades vidare med för att skapa en realtidslösning.

Ett fartygs rörelser, beskrivna i tre frihetsgrader med kontinuerlig dynamik, simulerades med hjälp av matematiska modeller av krafterna som verkar på fartyget. Ett icke-linjärt optimalt-styrningsproblem formulerades för varje scenario och löstes numeriskt med användning av diskretisering och en direkt multipelskjutningsmetod. Resultaten för det första problemet visade att fartyget kunde stanna utan större avvikelser från den förbestämda banan. Banor med olika krökning testades och det visades att bromssträckan blev något längre när krökningen på banan var större. Resultaten för det andra problemet visade att fartyget håller sig inom det säkra området och väljer en relativt rak väg för att stanna. Detta resulterar i en kortare tillryggalagd sträcka jämfört med lösningen av det första problemet.

Två olika realtidsansatser formulerades, dels en metod med receding-horizon och dels ett uppslagsbaserat tillvägagångssätt. Båda tillvägagångssätten löstes med realtidsgenomförbarhet, där tillväga-gångssättet med receding-horizon gav en bättre lösning medan det uppslagsbaserade tillvägagångssättet hade en kortare beräkningstid.

Nyckelord

Icke linjär optimering, olinjär optimal styrning, numerisk optimal styrning, realtidslösning, optimal bromsning, vikande horisont, modellering, fartygsmodell, propellerkrafter, roderkrafter, direkt multipelskjutningsmetod, diskretisering.

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Chapter 1

Introduction

In this master's thesis, a mathematical model of a vessel is used to simulate its behavior on water. An optimal control problem is then formulated to simulate the vessel coming to a full stop while sailing along a path. The thesis work was performed at ABB Corporate Research in Västerås in collaboration with KTH - Royal Institute of Technology.

1.1 Background

The transportation industry as a whole is rapidly changing, and it is largely due to high goals in automation. The shipping industry is certainly following this trend as companies work towards fully autonomous shipping. There are four levels of autonomy. The first level is where seafarers control the ship completely but have decision support from the autonomous system onshore. The second level is where the autonomous system controls the ship remotely from the shore but has crew members on board for backup. The third level is similar to the second, except that there are no crew members on board, and the fourth level corresponds to fully autonomous ships. When the ships are fully autonomous, the computers make all the decisions and the ship is monitored from shore (Wingrove 2020).

However, autonomous ships at any level are rare. There exist ships with advanced autonomous systems, such as Mayflower Autonomous Ship (Al-Heeti 2021), but there is a long way to go before all ships become autonomous. According to recent studies, some ships have autonomous systems at level two, but reaching level three, for

the general ship, is very difficult and level four is nowhere in sight (Brans 2021). Regulations are one of the more significant challenges of autonomous ships, largely due to safety concerns. However, autonomous shipping will have great advantages when the technology can show that it is as safe as the non-autonomous alternative (Wingrove 2020).

One of the advantages that will come with autonomous shipping is safety. In 2020 there were 2837 accidents and 38 fatalities reported within the European Union, and during 2014 – 2020, 89.5% of all occurrences were related to human error (European Maritime Safety Agency 2021). When removing the human aspect from sailing and replacing it with autonomous systems, these human-related occurrences will disappear.

One of the safety advantages of autonomous shipping is collision avoidance, where it is needed to stop the ship as fast as possible in order to avoid collision (Oneto et al. 2017). Stopping the vessel in the open sea with no obstacles is a challenging problem to begin with, and it becomes even more challenging when vessels are sailing in narrow spaces, for example, channels or archipelagos, since surrounding obstacles must be considered. This problem is challenging due to the complex relationship between the ship and its surroundings. The system consisting of the ship and its thrusters becomes complex because of the interaction with the water and the way the controllable forces are generated. On top of that, environmental factors, such as wind, waves, and ocean currents, have a significant impact on the motion of the ship, which complicates things even further (Fossen 2011).

Collision avoidance is not a planned activity, so the steps needed to stop the vessel can therefore not be determined ahead of time. Instead, it needs to be decided as soon as the danger of collision arises. The information on how the vessel would move to avoid collision must be available as fast as possible so that either an autonomous system or the crew has time to take appropriate actions (Li and Zheng 2020).

The movements of the vessel are generated with thrusters. Thrusters are units that generate thrust, for instance, by using propellers or rudder surfaces. There are multiple different types of thrusters that can be attached to a vessel, for example, azimuth thrusters. Azimuth thrusters are rotatable so that they can produce thrust in the direction needed (Sørensen 2011). There are different ways to stop a ship using azimuth thrusters. Jong-Yong et al. (2020) tested five different stopping procedures,

for instance, stopping the ship by inertia stopping, stopping by turning the azimuth thrusters 180 degrees, and stopping by changing the propeller direction. In the two latter methods, thrust is used to stop the ship, which minimized the braking distance.

In all the different methods, the vessel traveled some lateral distance. This lateral movement was not intentional but did not make a significant difference in testing the different methods. When surrounding obstacles need to be considered, both the lateral and forward motions must be accurate. In those conditions, the vessel may have access to less braking force, which could lead to longer stopping distances.

This leads to the research question for this project:

Considering a marine vessel with two rotating thrusters, can a real-time solution be implemented for an optimal braking problem along, or within a safe area of, a decided path?

1.2 Purpose

The purpose of this thesis is to know the final location of a vessel when it encounters critical situations in narrow spaces and needs to come to a complete stop. The calculations needed to get this information must be very fast, preferably within seconds, for either a crew member or a fully autonomous system, to make the right decisions in time.

1.3 Goals

The goals of this thesis are to:

1. Formulate an optimal braking problem along a path for a specific thruster configuration
2. Formulate a relaxed optimal braking problem where the path is replaced by a safe area
3. Develop an offline solution
4. Verify the offline solution in simulation

5. Develop a real-time solution
6. Verify the real-time solution in simulation

1.4 Thesis Boundaries and Limitations

For the main part of the thesis, environmental factors such as disturbances from wind, water currents, and other factors will be neglected to simplify the model.

The project will only focus on one type of thruster configuration. It will be a general model of two symmetrically mounted azimuth thrusters with rudder surfaces located at the stern of the vessel.

1.5 Methods

A literature study was done on several topics throughout the thesis. Maneuvering theory was studied to understand the motions of a vessel and how forces act on it. A new propeller model was defined to account for the desired movements of the vessel, but models of other forces and dynamics were used from literature. Optimal control theory was studied to formulate optimal control problems, and methods for implementing a real-time solution were also studied.

The optimal control problems, one on a path and another within a safe area, were implemented in MATLAB using CasADi and existing optimization packages. The movement of the vessel was also simulated in MATLAB to ensure that the solution from the optimal control problem acted in a reasonable way. An iterative process between the formulation of the optimal control problems and the implementation was used to ensure that the formulation suited the implementation and that the movement of the vessel was appropriate.

The real-time solution methods, a receding horizon method, and a lookup-based method were also formulated in MATLAB. In the former method, code generation was used along with the algorithm to speed up the computational time. In the latter, a database of optimal solutions was created and used in the algorithm.

1.6 Outline

The thesis will be outlined in the following way. Chapter 2 includes a description of the model of the ship and its thrusters. It also describes how the forces from the propeller and the rudder are modeled. Chapter 3 describes relevant optimal control theory. Chapter 4 introduces the optimal braking problem along a path and shows its results. Chapter 5 introduces the optimal braking within a safe area problem, explains the framework used for the problem, and shows its results. In Chapter 6, the real-time formulation is introduced and methods used to find the solution are described. Results are also shown at the end of the chapter. The final chapter, Chapter 7, includes a discussion about the thesis and an outline for future work.

Chapter 2

Models

This chapter describes a mathematical model for a surface vessel in three degrees of freedom. The vessel has two rotating thrusters, and a model of the force acting on the vessel from those thrusters will also be introduced. At the beginning of the chapter, the dynamics of the vessel is described, and then the model of the forces will be given, firstly, the propeller forces and thereafter the rudder forces. Lastly, the thruster dynamics will be introduced.

2.1 Ship Model

To describe the motion of a ship, 6 degrees of freedom (DOF) could be considered. These 6 DOFs are surge, sway, heave, roll, pitch, and yaw. In this thesis, environmental forces are neglected which means that it is assumed that the vessel is sailing in calm waters. The motion in heave, roll, and pitch can then be neglected and a model in 3 DOF will be used (Fossen 2011). The dynamics of the ship, which is split into two parts: kinematics and kinetics, can be described by

$$\dot{\boldsymbol{\eta}} = \mathbf{R}(\psi)\boldsymbol{\nu} \quad (\text{Kinematics}) \quad (2.1a)$$

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\eta}, \boldsymbol{\nu}) = \boldsymbol{\tau}(\boldsymbol{\eta}, \boldsymbol{\nu}) \quad (\text{Kinetics}) \quad (2.1b)$$

where $\mathbf{R}(\psi)$ is a rotation matrix, $\mathbf{M}$ is the mass inertia matrix, $\mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu}$ describes the coriolis forces, and $\mathbf{D}(\boldsymbol{\eta}, \boldsymbol{\nu})$ describes the damping force acting on the ship. The notation of the vectors are given by SNAME (1952) and is described further in Table 2.1.1. In this thesis, the model of the ships dynamics is given in Fossen (2011), and

Notation	Description
$\eta = [x \ y \ \psi]^T$	Position of the vessel given in Cartesian position (x, y) and heading angle ψ (n -frame)
$\nu = [u \ v \ r]^T$	Velocities of the vessel given in surge u , sway v and yaw r (b -frame)
$\tau = [X \ Y \ N]^T$	Forces affecting the vessel in surge X , sway Y , and yaw N (b -frame)

Table 2.1.1: Notation of vectors used to describe motion of a ship in 3 DOF.

the parameters for a supply vessel, which are available in the MSS toolbox (Perez et al. n.d.), is used.

A convenient practice when describing the motion of a ship is to define two reference frames, one body-fixed reference frame, and another inertial reference frame. Here, an Earth-fixed reference frame will be assumed to be inertial. The body-fixed reference frame, or b -frame, has its origin fixed to the center of origin of the vessel. The x_b -axis is directed from aft to fore, the y_b -axis is directed to the starboard, and the z_b is directed from top to bottom. The Earth-fixed frame, or n -frame, has its origin fixed to the surface of the Earth. The n -frame is often referred to as the North-East-Down coordinate system as x_n points North, y_n points East and z_n points down towards the center of the Earth (Fossen 2011). Figure 2.1.1 shows the relationship between the two reference frames and the generalized velocities in the b -frame.

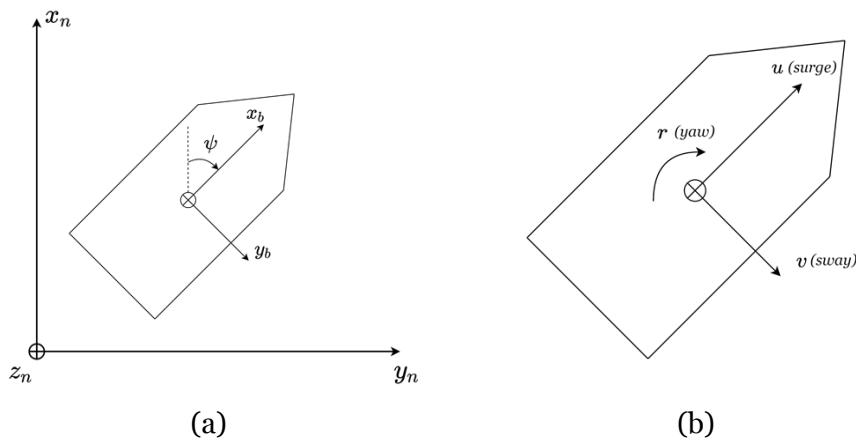


Figure 2.1.1: (a) shows the relationship between the two reference frames, n -frame and b -frame, and (b) shows the generalized velocities in the b -frame

2.1.1 Kinematics

The kinematics link together the generalized velocity $\boldsymbol{\nu}$ and the generalized position $\boldsymbol{\eta}$ of the vessel. It only describes the geometrical aspects of the vessel's motion without taking any forces into account. It can be described by

$$\dot{\boldsymbol{\eta}} = \mathbf{R}(\psi)\boldsymbol{\nu} \quad (2.2)$$

where $\mathbf{R}(\psi)$ is the rotation matrix given by

$$\mathbf{R}(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (2.3)$$

2.1.2 Kinetics

The kinetics describe the movement of the vessel as it is affected by external forces and moments. The relationship between the generalized velocity $\boldsymbol{\nu}$ and the generalized forces $\boldsymbol{\tau}$ affecting the motion of the ship can be described by

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\eta}, \boldsymbol{\nu}) = \boldsymbol{\tau}(\boldsymbol{\eta}, \boldsymbol{\nu}) \quad (2.4)$$

where the matrices are defined below using notation given by

citetsname. The forces, X_i , Y_i and N_i are derived with respect to the subscript i . For example, X_u is the force in the surge direction derived with respect to the velocity in the surge direction.

$\mathbf{M}$ is the inertia matrix given by

$$\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A = \begin{bmatrix} \bar{M} & 0 & 0 \\ 0 & \bar{M} & \bar{M}x_g \\ 0 & \bar{M}x_g & I_z + \bar{M}x_g^2 \end{bmatrix} - \begin{bmatrix} X_{\dot{u}} & 0 & 0 \\ 0 & Y_{\dot{v}} & Y_{\dot{r}} \\ 0 & Y_{\dot{r}} & N_{\dot{r}} \end{bmatrix} \quad (2.5)$$

where $\mathbf{M}_{RB}$ and $\mathbf{M}_A$ are the rigid-body inertia matrix and the added mass matrix respectively. $\bar{M}$ is the actual mass of the ship, x_g is the x coordinate of the center of gravity in b -frame, and I_z is the moments of inertia around the z_b axis.

The Coriolis matrix is given by

$$\mathbf{C}(\boldsymbol{\nu}) = \mathbf{C}_{RB}(\boldsymbol{\nu}) + \mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} 0 & -M_{11}r & -M_{23}r + (M_{11} - M_{22})v \\ M_{11}r & 0 & 0 \\ M_{23}r - (M_{11} - M_{22})v & 0 & 0 \end{bmatrix} \quad (2.6)$$

where $\mathbf{C}_{RB}(\boldsymbol{\nu})$ represents the centripetal and Coriolis terms, and $\mathbf{C}_A(\boldsymbol{\nu})$ is the added centripetal and Coriolis matrix. The notation M_{ij} corresponds to the element of $\mathbf{M}$ in the i^{th} row and j^{th} column.

Lastly, the damping term, $\mathbf{D}(\boldsymbol{\eta}, \boldsymbol{\nu})$, is given by

$$\mathbf{D}(\boldsymbol{\eta}, \boldsymbol{\nu}) = \mathbf{D}_L \boldsymbol{\nu} + \mathbf{d}_n(\boldsymbol{\nu}) = \begin{bmatrix} X_u & 0 & 0 \\ 0 & Y_v & Y_r \\ 0 & N_v & N_r \end{bmatrix} \boldsymbol{\nu} - \begin{bmatrix} X_{|u|u} & 0 & 0 & 0 & 0 \\ 0 & Y_{|v|v} & Y_{|v|r} & Y_{|r|v} & Y_{|r|r} \\ 0 & N_{|v|v} & N_{|v|r} & N_{|r|v} & N_{|r|r} \end{bmatrix} \begin{bmatrix} |u|u \\ |v|v \\ |v|r \\ |r|v \\ |r|r \end{bmatrix}. \quad (2.7)$$

2.2 Generalized forces

There are multiple different forces that act on a ship. These forces can be broken down into two main categories, controllable forces and forces caused by the environment. Here, these forces are assumed to be independent and the relationship is given by

$$\boldsymbol{\tau}(\boldsymbol{\eta}, \boldsymbol{\nu}) = \boldsymbol{\tau}_c(\boldsymbol{\eta}, \boldsymbol{\nu}) + \boldsymbol{\tau}_{\text{env}}(\boldsymbol{\eta}, \boldsymbol{\nu}) \quad (2.8)$$

where $\boldsymbol{\tau}(\boldsymbol{\eta}, \boldsymbol{\nu})$ is the total force acting on the ship, $\boldsymbol{\tau}_c(\boldsymbol{\eta}, \boldsymbol{\nu})$ is the controllable forces, and $\boldsymbol{\tau}_{\text{env}}(\boldsymbol{\eta}, \boldsymbol{\nu})$ is the environmental forces. In this thesis, the forces from the environment will be neglected for simplicity, i.e. $\boldsymbol{\tau}_{\text{env}}(\boldsymbol{\eta}, \boldsymbol{\nu}) = 0$.

The controllable forces come from the propeller and the rudder. These forces are assumed to be independent and can therefore be modeled separately and then combined linearly to get the complete force (Bergman et al. 2020). The model is given by

$$\boldsymbol{\tau}_c(\boldsymbol{\eta}, \boldsymbol{\nu}) = \boldsymbol{\tau}_p(\boldsymbol{\eta}, \boldsymbol{\nu}) + \boldsymbol{\tau}_r(\boldsymbol{\eta}, \boldsymbol{\nu}) \quad (2.9)$$

where $\tau_p(\eta, \nu)$ is the force from the propeller and $\tau_r(\eta, \nu)$ is the force from the rudder. The modelling of the controllable forces follows from the paper from Bergman et al. (2020).

2.2.1 Propeller model

The vessel considered in this thesis has two propellers that are a part of the rotatable thrusters. The thruster configuration can be seen in Figure 2.2.1. The propellers create force from their interaction with the water and can be controlled by their turning speed. To ensure that the model of the propeller force captures the correct behavior when sailing at high speeds, different models than the one from citereport4 were investigated. Smogeli (2006) describes a model using a sign-squared formula that accounts for high velocities of the vessel,

$$T_i = k_1 n_i |n_i| + k_2 U_{A,i} |n_i| + k_3 U_{A,i} |U_{A,i}| \quad (2.10)$$

where T_i is the total thrust, n_i is the propeller speed of propeller i , and $U_{A,i}$ is the speed of advance in the direction of propeller i . In this thesis, the propeller speed is assumed to be non-negative, but the speed of advance can be of either sign. This is because the propellers are rotating, and the speed of advance is always in the direction of the propeller. Thus even when the vessel is traveling at a positive velocity, the propellers can be turned in the opposite direction and then the speed of advance becomes negative. The model (2.10) is derived from a model of general thrust from a propeller defined by

$$T = K_T \rho |n| n D^4 \quad (2.11)$$

where ρ is the water density, D is the propeller diameter and K_T is the thrust coefficient. The thrust coefficient is given by

$$K_T = c_1 + c_2 J + c_3 |J| J \quad (2.12)$$

where $c_1 > 0$, $c_2 < 0$, and $c_3 < 0$ are constants given by van Lammeren, van Manen, and Oosterveld (1969). These constants are dependent on characteristics of the propeller blades, and the parameters used in this thesis can be seen in Table 2.2.1. The advanced ratio for thruster i is given by

$$J = \frac{U_{A,i}}{n_i D}. \quad (2.13)$$

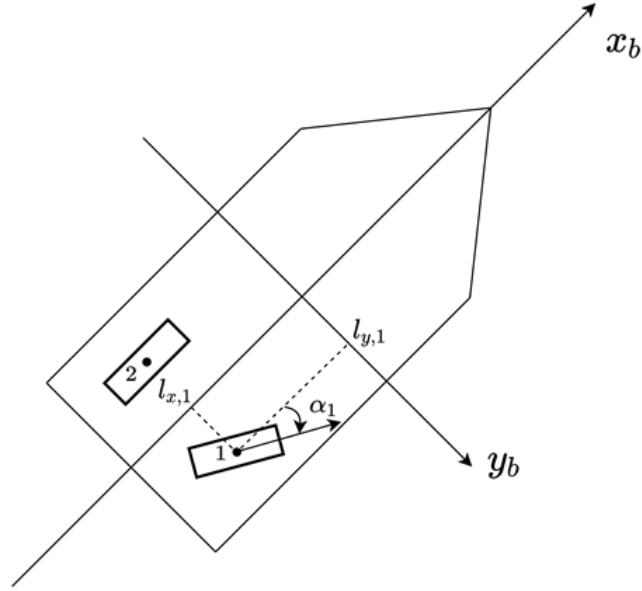


Figure 2.2.1: Thruster placement on the vessel. The thrusters are defined by their location, $l_{x,i}$ and $l_{y,i}$, and their angle α_i , with $i = 1, 2$

Parameter	Value
Pitch over diameter ratio	1
Blade area ratio	0.7
Number of blades	4
Propeller diameter	5

Table 2.2.1: Parameters related to the propellers used in this thesis

The velocity at thruster i is calculated according to

$$u_i = u - l_{y,i}r \quad (2.14a)$$

$$v_i = v + l_{x,i}r \quad (2.14b)$$

$$V_i = \sqrt{u_i^2 + v_i^2} \quad (2.14c)$$

where $l_{x,i}$ and $l_{y,i}$ are the positions in the x and y direction, respectively, of thruster i in b -frame. This velocity is used when calculating the speed of advance at propeller i , which is given by

$$U_{A,i} = V_i \cos(\alpha_i - \beta_i) \quad (2.15)$$

where the angle of the thrust is directed at angle α_i , and the angle of the velocity at the thruster is calculated as

$$\beta_i = \text{atan2}(v_i, u_i). \quad (2.16)$$

These derivations lead to the model

$$T_i = k_1 n_i |n_i| + k_2 U_{A,i} |n_i| + k_3 U_{A,i} |U_{A,i}|$$

defined in (2.10) where $k_1 = \rho D^4 c_1$, $k_2 = \rho D^3 c_2$, and $k_3 = \rho D^2 c_3$.

Even with only positive propeller speed, the propeller can still create negative force. This is important when the ship needs to slow down or come to a complete stop. When the propeller is creating negative force while keeping its propeller speed positive, the propeller is said to be *windmilling*. When the advance ratio J is high enough for the thrust coefficient K_T to equal zero, the thrust becomes negative. This happens as the speed of advance U_A is quite a lot higher than the propeller speed, and the thrust becomes negative while the torque, the rotational force, is still positive (Smogeli 2006).

The behavior of model (2.10) can be seen in Figure 2.2.2. As the speed of advance U_A becomes higher, the behavior starts to look like a parabola. It would be natural that as the propeller speed decreases that the force would also continue to decrease. This is not entirely true due to a phenomenon called stalling. Stalling is when the effective angle of attack of the propeller blade increases such that the force decreases. This happens when the speed of advance has a high magnitude, but the propeller speed is low. A model that includes stalling is the four-quadrant model first explained in

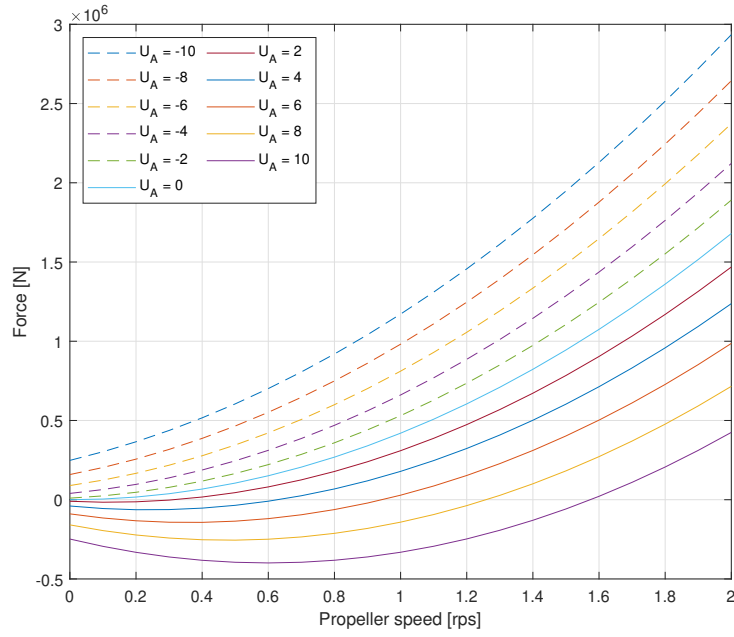


Figure 2.2.2: The force from a propeller using model (2.10)

Miniovich (1960). The behavior of that model can be seen in Figure 2.2.3. The four-quadrant model is included to show a more accurate behavior of a propeller force but is more complicated and will not be used in this thesis. When comparing the four-quadrant model with model (2.10), both models go from decreasing force magnitude to increasing force magnitude as the propeller speed increases. However, the four-quadrant model increases the force magnitude much earlier than model (2.10). This can be seen when the speed of advance is ten m/s, shown with the whole purple line in both graphs, as the four-quadrant model starts increasing the force magnitude just after 0.2 rps, while (2.10) does not start increasing the force magnitude until around 0.6 rps. Using (2.10) could therefore become problematic for the optimal solution when it needs slow propeller speed. An alternative from (2.10) will therefore also be looked at where that new model will be a non-decreasing function so that it is assumed that there is no stalling.

To create a non-decreasing function, as the propeller speed tends to zero, instead of the thrust increasing, a decrease in the thrust is needed. One way to do that is to flip the function around its minimum value. Using (2.10), the minimum value of the propeller speed, denoted with $n_{0,i}$, can be calculated. Using the fact that $U_{A,i}$ is constant for each

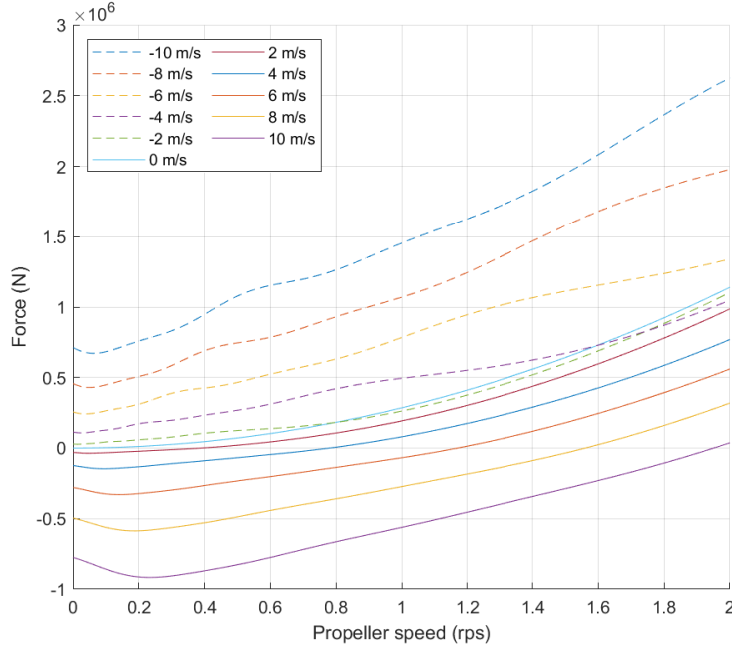


Figure 2.2.3: The four-quadrant model from Miniovich (1960), which shows a more accurate behavior. The decrease in magnitude when the propeller speed is close to zero is due to stalling.

time step, the derivative is set to zero to calculate the minimum, which is given by

$$n_{0,i} = \frac{-k_2 U_{A,i}}{2k_1}. \quad (2.17)$$

Now, let $T(n_i)$ denote the thrust at any point n_i and $T(n_{0,i})$ denote the thrust at the minimum point. Then, the function becomes

$$T_i = \begin{cases} T(n_i) & n_i \geq n_{0,i} \\ -T(n_i) + 2T(n_{0,i}) & n_i < n_{0,i} \end{cases}. \quad (2.18)$$

This is a non-decreasing function shown in Figure 2.2.4. This model shows a more accurate behaviour than (2.10) after 0.2 rpm. The problem with this model can be seen at 0 rpm, where if the propeller is not turning at all, there should be the same magnitude of force for the same magnitude in the speed of advance. At 0 rpm and ± 10 m/s, there should be a similar magnitude of force. Model (2.10) captures that behaviour better than this one. Now, the thrust along the propeller shaft has been modelled in two different ways, but the generalized force from both models can be written in the same way. The generalized force in b -frame follows from Bergman et al. (2020), where both

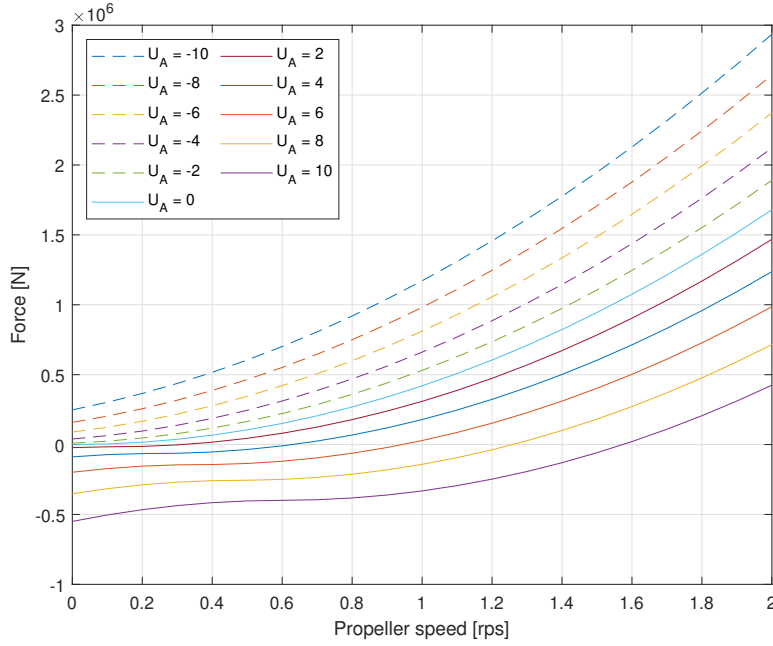


Figure 2.2.4: A nondecreasing model of thrust from a propeller shown in (2.18)

thrusters are assumed to be equivalent and the total force is the sum of the two. The total controllable force is now given by

$$\tau_p = \begin{bmatrix} \sum_{i=1}^2 T_i \cos(\alpha_i) \\ \sum_{i=1}^2 T_i \sin(\alpha_i) \\ \sum_{i=1}^2 T_i (l_{x,i} \sin(\alpha_i) - l_{y,i} \cos(\alpha_i)) \end{bmatrix} \quad (2.19)$$

where T_i can either be the force from the model in (2.10) or (2.18). The different models will be discussed further in Section 4.2.1.

2.2.2 Rudder model

The force from a rudder can be modeled according to Kang et al. (2008). By using the same assumption as the force from the propellers, the force from the two rudders can be combined linearly. The model of the total force from the rudders can be described by

$$\tau_r = \begin{bmatrix} \sum_{i=1}^2 -F_N(\alpha_i, \nu) \sin(\alpha_i) \\ \sum_{i=1}^2 F_N(\alpha_i, \nu) \cos(\alpha_i) \\ \sum_{i=1}^2 F_N(\alpha_i, \nu) (l_{x,i} \cos(\alpha_i) + l_{y,i} \sin(\alpha_i)) \end{bmatrix} \quad (2.20)$$

where $F_N(\alpha_i, \boldsymbol{\nu})$ describes the force at rudder i given by

$$F_N(\alpha_i, \boldsymbol{\nu}) = \frac{1}{2} \rho A_r F'_N(\alpha_i, \boldsymbol{\nu}) (u_i^2 + v_i^2) \quad (2.21)$$

where A_r is the effective rudder area and

$$F'_N(\alpha_i, \boldsymbol{\nu}) = C_N \sin(\alpha_i - \beta_i) \quad (2.22)$$

where C_N is a positive constant.

2.3 Thruster dynamics

The vessel is controlled with the two rotating thrusters attached towards the aft. The controllable components of the thrusters are the propeller speed and the thruster angle. Both of these components can be changed with time, and therefore the thruster dynamics will be introduced. To make the optimization problem easier to solve, simple approximations will be used. Here, the dynamics are simple integrators where the inputs can be limited. In that way, the rate of change in the angle can be bounded.

For the propeller speed n_i , the dynamics can be described with

$$\dot{n}_i = \Delta n_i \quad (2.23)$$

where Δn_i represents the rate of change of the propeller speed. Similarly the dynamics of the thruster angle α_i can be described with

$$\dot{\alpha}_i = \Delta \alpha_i \quad (2.24)$$

where $\Delta \alpha_i$ represents the rate of change in the thruster angle. Both (2.23) and (2.24) represent the change in the controls over time.

Chapter 3

Selected Topics in Optimal Control

Optimal control is a useful, systematic method to solve a variety of control problems, and its applications are widely used in engineering, science, and economics (Jönsson 2012). This approach can also be used for the optimization of trajectories and will be used in this thesis. This chapter will present selected topics in optimal control necessary to understand the remainder of the thesis. In the first section 3.1 the general formulation of the optimal control problem will be introduced. Section 3.2 will explain the discretization method used, and section 3.3 will introduce the direct multiple shooting method.

3.1 General formulation

The aim of an optimal control problem is to find a sequence of controls which minimizes the cost function given the system dynamics and boundary constraints. The cost function will generally consists of two terms and is given by

$$\phi(\mathbf{x}(t_f)) + \int_{t_i}^{t_f} f_0(t, \mathbf{x}(t), \mathbf{u}(t)) dt \quad (3.1)$$

where the first term is called the terminal cost, and penalizes the deviation from a desired final state. The second term is the integral term, or the cost associated with the state and control trajectories over a time interval, where the initial time and the final time are denoted t_i , and t_f , respectively. The initial time t_i is always assumed fixed while t_f can either be fixed or a variable. The system dynamics are defined in terms of a state-space equation $\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t))$, where the control variable is

restricted to a set of admissible controls $\mathbf{u}(t) \in U$ and the state vector is constrained to satisfy boundary conditions on a manifold $\mathbf{x}(t) \in S$. The initial and final manifold give additional constraints on the initial and final states, denoted S_i and S_f , respectively. Using this, the complete optimal control problem can now be formulated as

$$\text{minimize } \phi(\mathbf{x}(t_f)) + \int_{t_i}^{t_f} f_0(t, \mathbf{x}(t), \mathbf{u}(t)) dt \quad (3.2a)$$

$$\text{subject to } \dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t)) \quad (3.2b)$$

$$\mathbf{x}(t_i) \in S_i, \mathbf{x}(t_f) \in S_f \quad (3.2c)$$

$$\mathbf{x}(t) \in S \quad (3.2d)$$

$$\mathbf{u}(t) \in U. \quad (3.2e)$$

The solution to the optimal control problem is the admissible controls that transfer the state vector from a point on the initial manifold to a point on the final manifold while making the value of the cost function as small as possible (Jönsson 2012). This is, in general, a non-convex and nonlinear optimization problem.

3.2 Discretization method

In most cases, numerical methods are needed in order to solve an optimal control problem. Discretization is a common way of solving continuous problems numerically. The method splits the continuous-time interval into N discrete time steps. The length of a time step is defined as

$$h = t_f/N. \quad (3.3)$$

Now the states and the control can be discretized in the following way

$$\mathbf{x}(t_k) = \mathbf{x}_k, \quad \mathbf{u}(t_k) = \mathbf{u}_k \quad (3.4)$$

where

$$kh \leq t_k < (k+1)h, \quad k = 0, 1, \dots, N-1. \quad (3.5)$$

Using the discretization, the optimal control problem now becomes

$$\text{minimize} \quad \phi(\mathbf{x}_N) + \sum_{k=0}^{N-1} h f_0(kh, \mathbf{x}_k, \mathbf{u}_k) \quad (3.6a)$$

$$\text{subject to} \quad \mathbf{x}_{k+1} = \mathbf{F}(kh, \mathbf{x}_k, \mathbf{u}_k) \quad (3.6b)$$

$$\mathbf{x}_0 \in S_i, \mathbf{x}_N \in S_f \quad (3.6c)$$

$$\mathbf{x}_1, \dots, \mathbf{x}_{N-1} \in S \quad (3.6d)$$

$$\mathbf{u}_k \in U. \quad (3.6e)$$

The discretized dynamics are given as the function $\mathbf{F}(kh, \mathbf{x}_k, \mathbf{u}_k)$. Different methods can be used to approximate the dynamics after the states have been discretized. One of those methods is the 4th order Runge-Kutta method, which estimates the slope for each interval by calculating the slope at four different points within the interval (Diehl 2011). For a general function $\dot{\mathbf{x}} = \mathbf{f}(t_k, \mathbf{x}_k, \mathbf{u}_k)$ and a step size h , the method is

$$\mathbf{k}_1 = \mathbf{f}(t_k, \mathbf{x}_k, \mathbf{u}_k) \quad (3.7a)$$

$$\mathbf{k}_2 = \mathbf{f}\left(t_k + \frac{h}{2}, \mathbf{x}_k + \frac{h}{2}\mathbf{k}_1, \mathbf{u}_k\right) \quad (3.7b)$$

$$\mathbf{k}_3 = \mathbf{f}\left(t_k + \frac{h}{2}, \mathbf{x}_k + \frac{h}{2}\mathbf{k}_2, \mathbf{u}_k\right) \quad (3.7c)$$

$$\mathbf{k}_4 = \mathbf{f}(t_k + h, \mathbf{x}_k + h\mathbf{k}_3, \mathbf{u}_k) \quad (3.7d)$$

$$\mathbf{F}(t_k, \mathbf{x}_k, \mathbf{u}_k) = \mathbf{x}_k + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \quad (3.7e)$$

where now the discrete dynamics are defined as $\mathbf{x}_{k+1} = \mathbf{F}(t_k, \mathbf{x}_k, \mathbf{u}_k)$. This 4th order Runge-Kutta method is the one that will be used in this thesis to approximate the discretization of the continuous terms in the formulation, the integral term in the cost function.

3.3 Direct multiple shooting method

A popular method for solving optimal control problems is the direct multiple shooting method (Diehl 2011). The method can be used after the problem has been discretized into smaller intervals. The method is based on solving the problem for each of the smaller intervals and then making sure the solution is continuous on the whole interval by introducing continuity constraints. The method can be described in the following

way:

For each discretized interval $t \in [t_k, t_{k+1}]$, there is a corresponding $\mathbf{x}_k$ and $\mathbf{u}_k$, and an artificial initial value $\mathbf{s}_k$ such that

$$\dot{\mathbf{x}}_k(t) = \mathbf{f}(\mathbf{x}_k(t), \mathbf{u}_k), \quad t \in [t_k, t_{k+1}] \quad (3.8a)$$

$$\mathbf{x}_k(t_k) = \mathbf{s}_k. \quad (3.8b)$$

In the first iteration, $\mathbf{s}_k$ is initialized, for instance, by an initial guess of the states. This gives trajectory pieces $\mathbf{x}_k(t; \mathbf{s}_k, \mathbf{u}_k)$ for each interval, where the semicolon means that the trajectories are determined by the following values. Now, the optimization problem

$$\text{minimize} \quad \phi(\mathbf{s}_N) + \sum_{k=0}^{N-1} f_{0,k}(\mathbf{s}_k, \mathbf{u}_k) \quad (3.9a)$$

$$\text{subject to} \quad \mathbf{s}_0 = \mathbf{x}_0 \quad (\text{initial value}) \quad (3.9b)$$

$$\mathbf{x}_k(t; \mathbf{s}_k, \mathbf{u}_k) = \mathbf{s}_{k+1} \quad (\text{continuity}) \quad (3.9c)$$

$$\mathbf{x}_1, \dots, \mathbf{x}_{N-1} \in S \quad (\text{path constraints}) \quad (3.9d)$$

$$\mathbf{x}_N \in S_f \quad (\text{terminal constraint}) \quad (3.9e)$$

$$\mathbf{u}_k \in U \quad (\text{controls constraint}) \quad (3.9f)$$

can be solved by numerical optimization. Since the method chooses artificial initial values on each interval, (3.9c) is important to make sure that the solution to the problem is continuous over all the intervals.

Multiple shooting is a good method for nonlinear optimization problems because it is not more computationally heavy than other methods and can use existing forward solvers on each subinterval without relying heavily on the number of subintervals. Since the optimization problem is solved in each subinterval, it is also better at dealing with unstable nonlinear systems than other similar methods (Diehl 2011).

Chapter 4

Optimal Braking Along a Path

In this chapter, an optimal braking problem along a path is formulated. The goal of the chapter is to find the optimal way to stop a vessel while staying as close as possible to a pre-decided path. This chapter will present the problem formulation in Section 4.1, and in Section 4.2, results from the optimal control problem are given.

4.1 Problem formulation

Using the general formulation from Chapter 3.1, an optimal control problem for optimal braking can be formulated. The state vector $\mathbf{x}(t)$ is composed of the generalized position of the vessel $\boldsymbol{\eta}$, the generalized velocities $\boldsymbol{\nu}$, and the control signals. The position and velocities are defined in Chapter 2.1 and the control signals are given by $\begin{bmatrix} n_1 & \alpha_1 & n_2 & \alpha_2 \end{bmatrix}^T$ where n_i is the propeller speed of propeller i , and α_i is the angle of thruster i . The extended state vector is given by

$$\mathbf{x}(t) = \begin{bmatrix} \boldsymbol{\eta}^T & \boldsymbol{\nu}^T & n_1 & \alpha_1 & n_2 & \alpha_2 \end{bmatrix}^T = \begin{bmatrix} x & y & \psi & u & v & r & n_1 & \alpha_1 & n_2 & \alpha_2 \end{bmatrix}^T. \quad (4.1)$$

The control vector is here denoted by $\mathbf{u}(t)$, and represents the change in the control signals. The control vector is given by

$$\mathbf{u}(t) = \begin{bmatrix} \Delta n_1 \\ \Delta \alpha_1 \\ \Delta n_2 \\ \Delta \alpha_2 \end{bmatrix}. \quad (4.2)$$

The system dynamics is given by

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{\eta} \\ \dot{\nu} \\ \dot{n}_1 \\ \dot{\alpha}_1 \\ \dot{n}_2 \\ \dot{\alpha}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{R}(\psi)\boldsymbol{\nu} \\ \mathbf{M}^{-1}(\boldsymbol{\tau} - \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} - \mathbf{D}(\boldsymbol{\eta}, \boldsymbol{\nu})) \\ \Delta n_1 \\ \Delta \alpha_1 \\ \Delta n_2 \\ \Delta \alpha_2 \end{bmatrix} = \mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t)) \quad (4.3)$$

where the dynamics of the position and velocity of the vessel are described in Section 2.1 and the dynamics of the control signals are described in Section 2.3.

The cost function is constructed such that it should minimize the final time, the change in control signals, the value of the propeller speed, the total distance traveled, and the distance from the decided path. Both of the distance terms must be calculated at each step after discretization and will thus be added in the discretized formulation of the problem, while the other terms are added to the continuous integral term. Without loss of generality it is possible to set the initial time to zero, thus $t_i = 0$, and then the final time is considered a free variable. The final time will then be the only term in the terminal cost, and the other terms are a part of the integral term,

$$\phi(\mathbf{x}(t_f)) = 10t_f \quad (4.4a)$$

$$f_0(t, \mathbf{x}(t), \mathbf{u}(t)) = 10(\Delta n_1^2 + \Delta n_2^2) + 10(\Delta \alpha_1^2 + \Delta \alpha_2^2) + n_1^2 + n_2^2. \quad (4.4b)$$

This continuous formulation of the dynamics and the cost function are discretized using the methods described in Chapter 3.2 which results in this formulation of the optimal control problem,

$$\begin{aligned} \text{minimize} \quad & \phi(\mathbf{x}_N) + \sum_{k=0}^{N-1} h f_0(kh, \mathbf{x}_k, \mathbf{u}_k) + 100(\hat{\alpha}_{1,k}(\mathbf{x}_k) + \hat{\alpha}_{2,k}(\mathbf{x}_k)) + 300\mathbf{d}_{\text{path},k}(\mathbf{x}_k)^2 + 3\mathbf{d}_{\text{total},k}(\mathbf{x}_k) \\ & \quad (4.5a) \end{aligned}$$

$$\text{subject to} \quad \mathbf{x}_{k+1} = \mathbf{F}(kh, \mathbf{x}_k, \mathbf{u}_k) \quad (4.5b)$$

$$\mathbf{x}_0 \in S_i, \mathbf{x}_N \in S_f \quad (4.5c)$$

$$\mathbf{x}_1, \dots, \mathbf{x}_{N-1} \in S \quad (4.5d)$$

$$\mathbf{u}_k \in U \quad (4.5e)$$

where $F(kh, \mathbf{x}_k, \mathbf{u}_k)$ represent the discretized dynamics, $f_0(kh, \mathbf{x}_k, \mathbf{u}_k)$ represents the discretized integral part of the cost function, and the terminal cost is still the same, $\phi(\mathbf{x}_N) = t_f = hN$. Since t_f is a variable and N is fixed, h is not fixed in the optimization problem.

The distance terms have been added to the discretized formulation. The $d_{\text{path},k}(\mathbf{x}_k)$ represent the distance from the position of the vessel to the closest point on the path. This term requires the vessel to stay close to the path to minimize the cost. The distance is given by

$$d_{\text{path},k}(\mathbf{x}_k) = \sqrt{(x_p - x_k)^2 + (y_p - y_k)^2 + (\psi_p - \psi_k)^2} \quad k = 1, 2, \dots, N \quad (4.6)$$

where (x_p, y_p) is the point on the path that is the closest to the position of the vessel at time k . The path is created with discrete data points, retrieved by simulating the motion of the vessel and then using linear interpolation to connect the data points. Since linear interpolation is used to create the path, a projection from the position of the vessel onto the closest line segment can be used to calculate the distance from the path. The rotational position ψ_p of the ship at the projected point is calculated as a linear combination of the angle at the endpoints of the line segment.

The $d_{\text{total},k}(\mathbf{x}_k)$ is the total distance the vessel travels while braking. This term penalizes the distance that the vessel is travelling, to try and travel the shortest distance possible, as that is the goal of the problem. This term is calculated as

$$d_{\text{total},k}(\mathbf{x}_k) = \sqrt{(x_{k+1} - x_k)^2 + (y_{k+1} - y_k)^2}. \quad (4.7)$$

The terms $\hat{\alpha}_{1,k}(\mathbf{x}_k)$ and $\hat{\alpha}_{2,k}(\mathbf{x}_k)$ are to prefer the inwards position of the thrusters. They are defined as

$$\hat{\alpha}_{1,k}(\mathbf{x}_k) = \begin{cases} \alpha_1^2 & \text{if } \alpha_1 > 0 \\ 0 & \text{else} \end{cases} \quad (4.8a)$$

$$\hat{\alpha}_{2,k}(\mathbf{x}_k) = \begin{cases} \alpha_2^2 & \text{if } \alpha_2 < 0 \\ 0 & \text{else} \end{cases}. \quad (4.8b)$$

In Section 4.1.1, constraints are introduced to limit the outward position so that the thrusters do not flush water onto each other. This part of the cost function is to help

the thrusters start with an inward position and eliminates the problem of getting stuck in a local minimum where an outwards position is chosen at the beginning of the solution.

4.1.1 Constraints

Both the state vector and the control vector are limited by a set of constraints. The set of constraints for the state vector can be described with manifolds, and the set of constraints for the controls is a set U called the admissible controls. The goal of the optimal control problem is to find admissible controls that enable the problem to travel from an initial manifold S_i to a terminal manifold S_f , with the lowest cost possible. Additional constraints are also applied to the state vector at each step of the problem, which can be described in the set S .

The initial and terminal manifold for the problem can be described as

$$S_i = \{\mathbf{x} \in \mathbb{R}^{10} | x = x_i, y = y_i, \psi = \psi_i, u = u_i, v = v_i, r = r_i, n_1 = n_2 = n_i, \alpha_1 = \alpha_2 = \alpha_i\} \quad (4.9a)$$

$$S_f = \{\mathbf{x} \in \mathbb{R}^{10} | u = v = r = 0, n_1 = n_2 = 0.2, \alpha_1 = -90^\circ, \alpha_2 = 90^\circ\} \quad (4.9b)$$

respectively. In this chapter, the vessel will start at the origin with velocity of 10 m/s in surge direction, and the thrusters will have different angles depending on the path they are travelling along. The sway velocity, yaw rate and propeller speed will have an initial value which keeps the vessel at 10 m/s at the initial angle. The formulation works in general since the vessel's starting position can always be transformed to the origin. The terminal manifold describes the state of the vessel when it has stopped completely. This corresponds to all of the velocities are at 0, and the thrusters are in what is called a neutral position when they are turned inwards 90° and have a low propeller speed at 0.2 rps. The final position of the vessel is completely free and thus no limitations are set on them in the manifold.

For each step between the the initial and terminal manifold, there are constraints on the values of the state vectors which form the set S . In this case, there are lower and

upper bounds, denoted by $\mathbf{x}_{lb}$ and $\mathbf{x}_{ub}$, respectively, and are given by

$$\mathbf{x}_{lb} = \begin{bmatrix} -\infty & -\infty & -\infty & 0 & -\infty & -\infty & 0.2 & -\infty^\circ & 0.2 & -\infty^\circ \end{bmatrix}^T \quad (4.10a)$$

$$\mathbf{x}_{ub} = \begin{bmatrix} \infty & \infty & \infty & \infty & \infty & \infty & 2 & \infty^\circ & 2 & \infty^\circ \end{bmatrix}^T \quad (4.10b)$$

and thus the state vector is bounded

$$\mathbf{x}_{lb} \leq \mathbf{x}_k \leq \mathbf{x}_{ub} \quad k = 1, 2, \dots, N. \quad (4.11)$$

The position of the vessel is completely free, and so are the velocities, except the surge velocity must be non-negative. This ensures that the vessel does not travel backward before stopping. The propeller speed is constrained between 0.2 rps and 2 rps. The angle of the thrusters is free in the upper and lower bounds, but additional constraints are set on the angle depending on the total velocity of the vessel, which is described below

$$\alpha_1 = \begin{cases} -30 < \alpha_1 < 30 & \text{if } V \geq 5 \\ -240 < \alpha_1 < 60 & \text{else} \end{cases} \quad (4.12a)$$

$$\alpha_2 = \begin{cases} -30 < \alpha_2 < 30 & \text{if } V \geq 5 \\ -60 < \alpha_2 < 240 & \text{else} \end{cases} \quad (4.12b)$$

where $V = \sqrt{u^2 + v^2}$ denotes the total velocity of the vessel. If the velocity of the vessel is more than 5 m/s, the angle of the thrusters can not be more than 30° in either direction. These constraints are in place to make sure that the thrusters will not be damaged due to the force of the water. If the velocity is less than 5 m/s, the thrusters can turn more freely. They are, however, restricted such that they can only turn 60° outwards, but they are not restricted when turning inwards unless they turn a whole round towards the outwards position, which they are then limited to 240° . These limits are introduced to make sure that the thrusters do not flush water onto the other thruster.

To get differentiable constraints, (4.12) is relaxed using the function (4.14), where (4.13) is the sigmoid function. The functions are given by

$$f(V) = \frac{1}{1 + e^{c(V-v_0)}} \quad (4.13)$$

where c is the sigmoid constant denotes how steep the change from the lower and upper bounds is, and v_0 denotes the constraining velocity which in this case $v_0 = 5$, and

$$s = s_1 + (s_2 - s_1)f(V) \quad (4.14)$$

where s denotes value of the constraint, s_2 denotes the value of the if statement, and s_1 denotes the value of the else statement. This function is used for the upper and lower bounds for each thruster. For example for the upper bound of thruster two, $s = 240 + (30 - 240)f(V)$. Given the constant c and the velocity of the vessel the constraints can be calculated.

There are also constraints on the values of the control vector. Similarly to the state vector, the control vector has constraints at each step, which is an upper and lower bound on the controls. Since the control vector denotes the change in controls, the constraints represent how abruptly the controls can change in each step. The limit in the controls are given by

$$\mathbf{u}_{lb} = \begin{bmatrix} -0.08 & -7.2 & -0.08 & -7.2 \end{bmatrix}^T \quad (4.15a)$$

$$\mathbf{u}_{ub} = \begin{bmatrix} 0.08 & 7.2 & 0.08 & 7.2 \end{bmatrix}^T \quad (4.15b)$$

$$\mathbf{u}_{lb} \leq \mathbf{u}_k \leq \mathbf{u}_{ub} \quad k = 0, 1, 2, \dots, N - 1 \quad (4.15c)$$

where the bound on the change of propeller speed is given in rps/s, and the bound on the change of angle is given in degrees per second.

4.2 Results

The optimal braking problem (4.5) from the formulation above was solved numerically by using the direct multiple shooting method. The problem was implemented in MATLAB using CasADi (Andersson et al. 2019) and solved using IPOPT (Wächter and L.Biegler 2006) with MUMPS (Amestoy et al. 2001). The main results showed that the vessel traveled and stopped very close to the path, given different curvatures and directions of the path. As the path direction and curvature increased, the distance traveled and the final time increased.

4.2.1 Propeller model

In Chapter 2, two different models of the propeller force were proposed. The goal was to have a simple model that gives similar behavior to the four-quadrant model shown in Figure 2.2.3. The two different models can be seen in Figures 4.2.1 and Figure 4.2.2. Figure 4.2.1 shows a solution using model (2.10), and Figure 4.2.2 shows a solution using the non-decreasing model (2.18). The solutions are obtained by using the problem formulation above.

From the formulation of the problem, it is known that the propeller speed is constrained between 0.2 rps and 2 rps. When looking back at the four-quadrant model, the stalling occurs between 0 rps and 0.2 rps. The stalling is therefore not relevant within the allowed propeller speed range. This behavior is captured with the non-decreasing model (2.18) but not with the model (2.10). In (2.10) the magnitude of the force is still decreasing after 0.2 rps, and for the highest speed, it decreases until 0.6 rps. This prolonged decreasing behavior has an impact on the solution, which is shown in Figures 4.2.1 and 4.2.2. In the non-decreasing model (2.18), the propeller speed goes all the way to minimum propeller speed, while (2.10) does not. The only reason why (2.10) does not reach the minimum propeller speed is because, according to the model, doing that would decrease the braking force.

For this reason, the non-decreasing model (2.18) shows a more correct behavior within the allowed propeller speeds when compared to the four-quadrant model. Thus (2.18) has been chosen for all further results in this thesis.

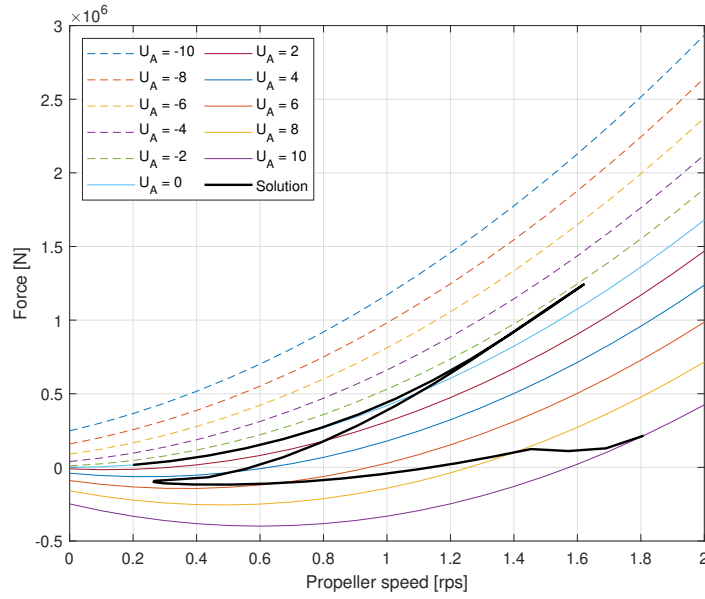


Figure 4.2.1: A solution when using model (2.10). The solution starts where propeller speed is at 1.5 rps and the speed of advance is 10 m/s, and ends at 0.2 rps and 0 m/s. It shows the force generated in the direction of the propeller given the propeller speed. In the beginning, the solution starts to go towards the minimum propeller speed but stops before the increase in force. It then increases the propeller speed because at that time, the thrusters have turned around almost 180°, so it is creating force in the opposite direction of the direction the vessel is traveling in.

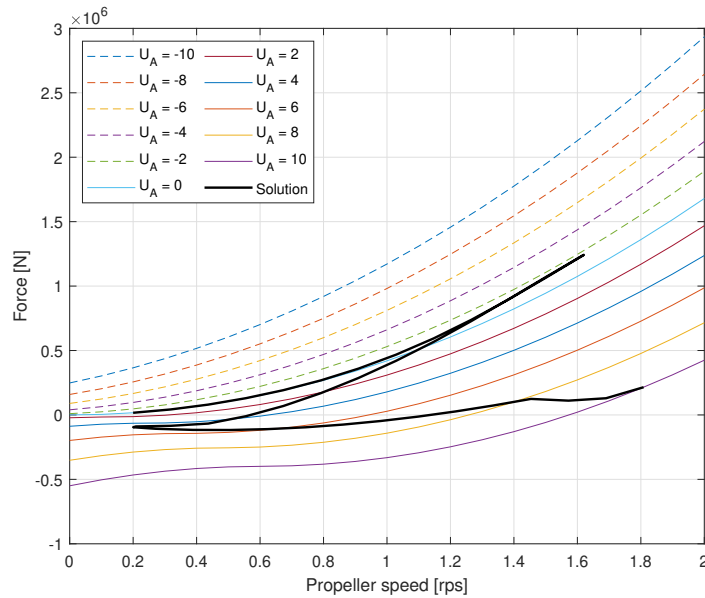


Figure 4.2.2: A solution when using model (2.18). This solution starts where the propeller speed is at 1.5 rps and the speed of advance is 10 m/s, and ends at 0.2 rps and 0 m/s, and also shows the force generated in the direction of the propeller given the propeller speed. Since there is no increase of force as the solution goes towards lower rps, the solution will reach the minimum propeller speed before the thrusters have turned 180°, which shows more similar behavior to the four-quadrant model.

4.2.2 Straight path

Simulating the movements of the vessel on a straight path is shown first. The path is simulated with the thrusters at 0 degrees, which makes the initial value of the thrusters $\alpha_1 = \alpha_2 = 0^\circ$. The velocities are given as $u = 10$ m/s, $v = 0$ m/s, and $r = 0$ rad/s, and the propeller speed is chosen such that the vessel travels at a steady-state, $n_1 = n_2 \approx 1.81$ rps. This is considered a good baseline to see the effects of braking on a curved path, as stopping on a straight line is considered an easier case. In Figure 4.2.3, the traveled path and the orientation of the ship can be seen. The vessel stays on the path completely until it stops. It travels a total of 256.0 meters and takes 59.2 seconds to come to a full stop.

In Figure 4.2.4, the change in the generalized position and velocity over time can be seen. The distance in the x direction increases with time, and the surge velocity decreases as expected. The other states have negligible changes close to zero and do not affect how the vessel moves along the path. The control signals can be seen in Figure 4.2.5. The controls are symmetrical, which means that both thrusters are changing in the same way at exactly the same time. This allows the vessel to continue on the straight path without changing its way.

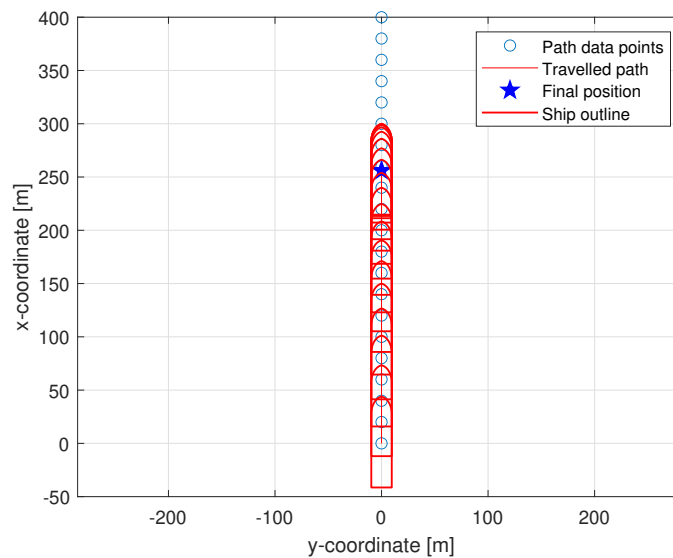


Figure 4.2.3: A simulation of the vessel braking on a straight path. The total distance travelled is 256.0 meters, and it takes 59.2 seconds for the ship to come to a complete stop.

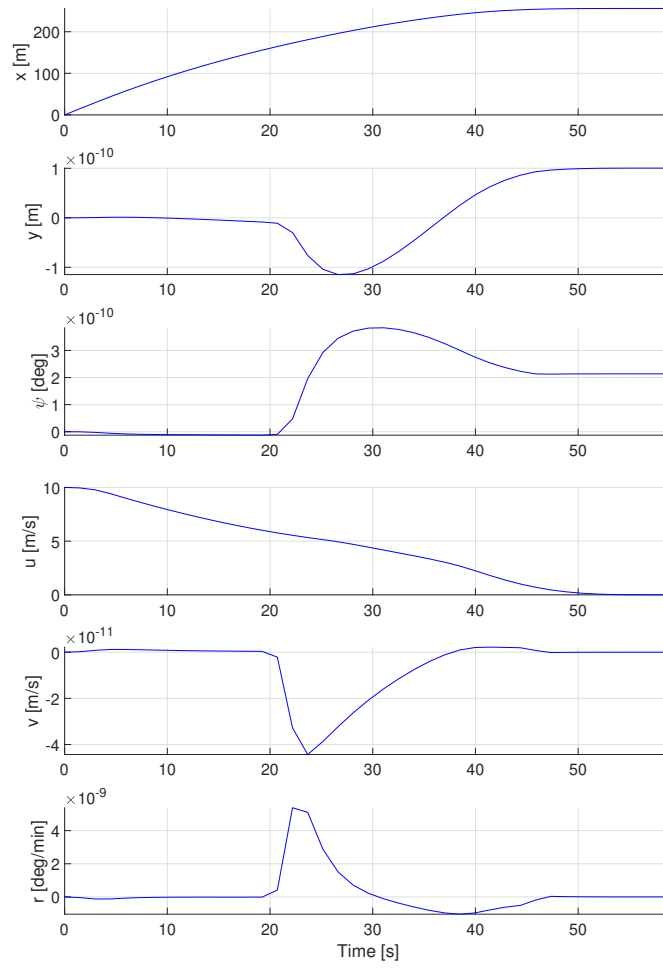


Figure 4.2.4: The plots show the position and velocity of the vessel over time when braking along a straight path. It is worth noting that the states y , ψ , v , and r have small changes that are negligible.

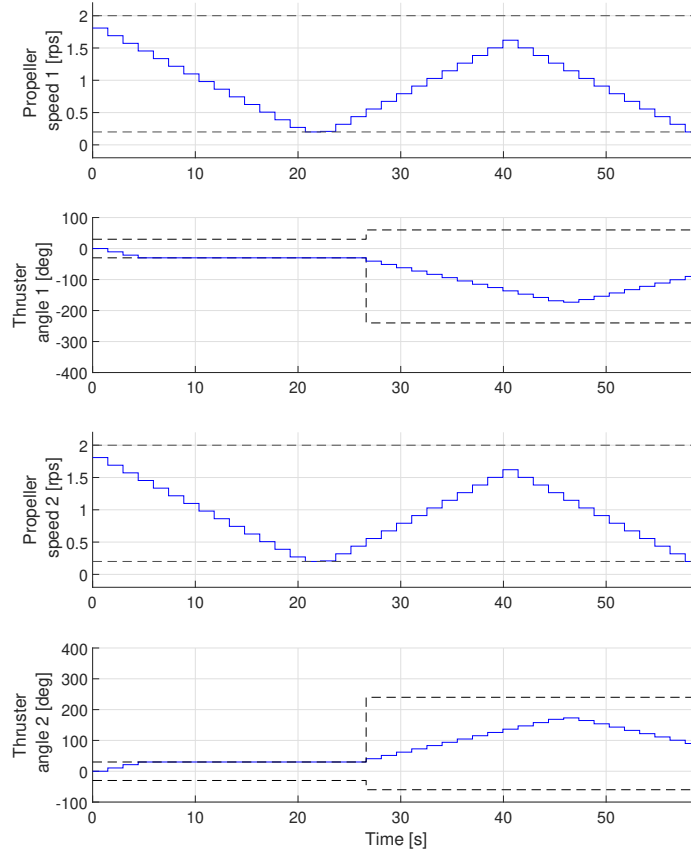


Figure 4.2.5: The plots show the control signals of the thrusters when braking along a straight path. The thruster angles are symmetric and the propeller speed is the same for both propeller. The black dotted lines show the constraints on the turning angle of the thrusters, and the constraints on the propeller speed.

4.2.3 Curved line

Now a curved path was simulated using the dynamics of the vessel, where the angles were 10 degrees on both thrusters. To correspond to the curved path, the initial state of the angle of the thrusters was also 10 degrees. The velocities are given as $u = 10$ m/s, $v \approx 1.11$ m/s, and $r \approx -0.016$ rad/s, and the propeller speed is chosen such that the vessel travels at a steady-state, $n_1 = n_2 \approx 1.89$ rps, i.e. the steady-state with 10 degree thruster angles. As shown in Figure 4.2.6, the vessel travels and stops on the path. The orientation of the ship is also ideal as it does not deviate too much from the heading of the original path. Since the path now has more curvature, the vessel travels further before it comes to a stop, 306.4 meters, which is almost 70 more meters than on a straight path. It also takes a little bit longer to come to a full stop, as it takes 65.7

seconds, almost 10 seconds longer than on the straight path.

In Figures 4.2.7 and 4.2.8, the states and controls over time can be seen. Naturally, there are more movements now than when the path was straight, but that was to be expected as now the vessel is moving in both forward and sideways directions and also rotates. With that, there is also more velocity in the sideways and rotational movements. The control signals also change a bit, as they are not as symmetrical anymore. There is still some symmetry, but to move along the path, the thrusters can not move in the same way. Otherwise, the vessel would move straight. When comparing the states and the control signals of the curved path with the ones of the straight path, the reason why the vessel travels further along the curved path can be seen. It is possible to look at the surge velocity as it lowers slightly slower for the curved path. However, the biggest difference can be seen in the thrust angle constraints. The constraints open up once the velocity of the vessel is below 5 m/s. In the curved path solution, one can see that the constraints open up almost 10 seconds after they do in the straight path solution since less negative force can be created by the thrusters. As the vessel is turning along with the path, some of the force must go into steering the ship. This indicates that the vessel is lowering its velocity slower and thus cannot start turning the thrusters as quickly. The vessel is therefore traveling for longer at high speeds, which explains the further distance traveled. It is worth noting that the initial sway and yaw velocities are higher than in the straight line case, which increases the total velocity of the vessel, and could also be a factor in why the vessel takes longer to decrease its velocity, along with the factor that it needs to stay on the path.

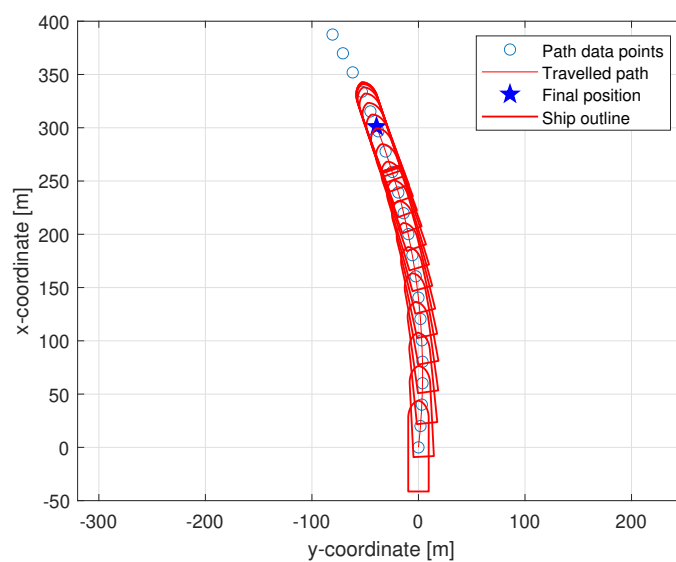


Figure 4.2.6: A simulation of the vessel braking on a curved path which was simulated with a 10 degree angle on both thrusters. The total distance travelled is 306.4 meters, and it takes 65.7 seconds for the ship to come to a complete stop.

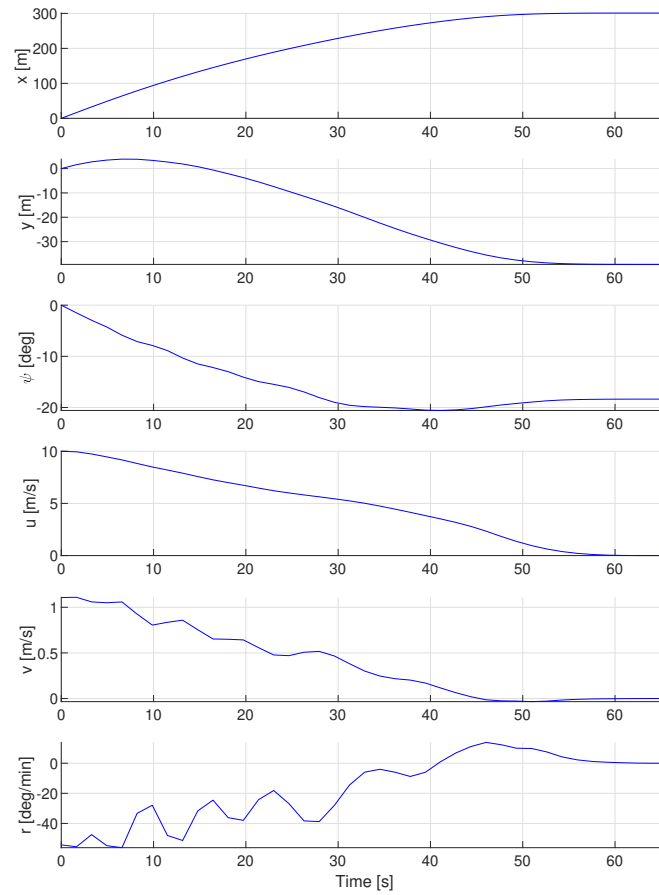


Figure 4.2.7: The plots show the position and velocity of the vessel over time when braking along the curved path. Now, there is more movement in sway and yaw, which allows the vessel to turn along the path. The surge velocity is also decreasing slower than when braking on a straight line.

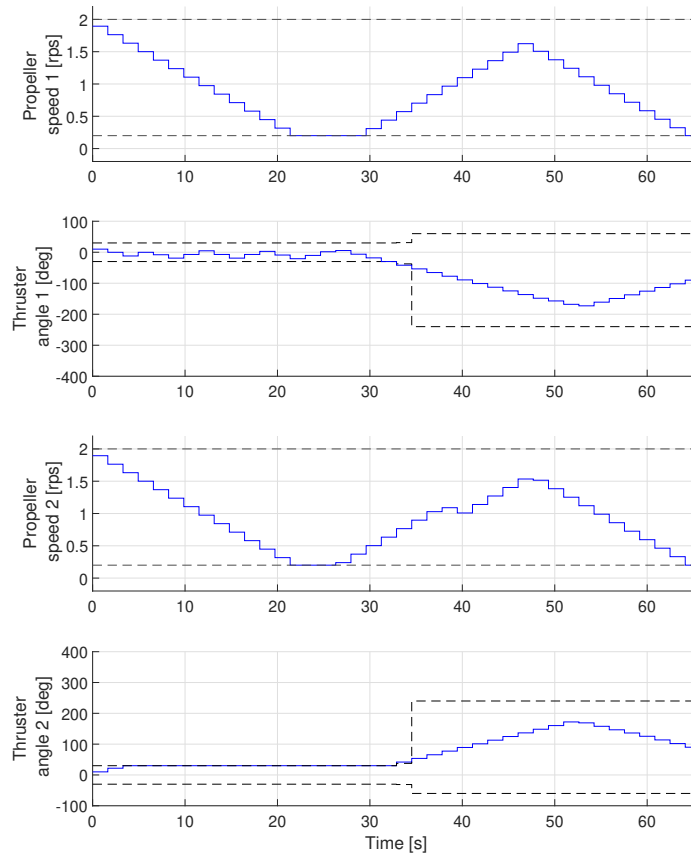


Figure 4.2.8: The plots show the control signals of the thrusters when braking along the curved path. There is not as much symmetry in this solution compared to the straight path solution. The black dotted lines show the constraints on the turning angle of the thrusters and the constraints on the propeller speed.

Chapter 5

Optimal Braking Within a Safe Area

This chapter formulates the optimal control problem a little differently than in Chapter 4. Instead of braking along a path, a safe area is created such that the vessel should brake and stay within the area. This chapter will present the problem formulation in Section 5.1 and the results in Section 5.2.

5.1 Problem formulation

There are only small changes to the problem formulation of the optimal braking problem when requiring the vessel to stay within a safe area instead of following a path. The state vector, the control vector and the system dynamics are the same, given in (4.1), (4.2), and (4.3), respectively. The continuous cost function, given in (4.4), does not change either, but the cost function in the formulation on the optimal control problem changes slightly. The current optimal control problem is given by

$$\begin{aligned} \text{minimize} \quad & \phi(\mathbf{x}_N) + \sum_{k=0}^{N-1} h f_0(kh, \mathbf{x}_k, \mathbf{u}_k) + 100(\hat{\alpha}_{1,k}(\mathbf{x}_k) + \hat{\alpha}_{2,k}(\mathbf{x}_k)) + \mathbf{d}_{\text{total},k}(\mathbf{x}_k) \end{aligned} \quad (5.1a)$$

$$\text{subject to} \quad \mathbf{x}_{k+1} = \mathbf{F}(kh, \mathbf{x}_k, \mathbf{u}_k) \quad (5.1b)$$

$$\mathbf{x}_0 \in S_i, \mathbf{x}_N \in S_f \quad (5.1c)$$

$$\mathbf{x}_1, \dots, \mathbf{x}_{N-1} \in S \quad (5.1d)$$

$$\mathbf{u}_k \in U. \quad (5.1e)$$

The change in the cost function of (5.1) is now that the term for the distance from the path is not needed anymore, but the total distance is, of course, still important to minimize. There still exists a path that the vessel is sailing along, but once it has to stop, it does not matter anymore whether the vessel stops close to the path as long as it stays within the safe area. Along with the constraints mentioned in Section 4.1.1, making sure that the vessel stays within a safe area is added as an additional constraint, mentioned in more detail in Section 5.1.1.

5.1.1 Safe area and obstacles

The safe area is built from the path that the vessel is sailing along. The path is generated by simulating the movements of the vessel and retrieving sample points along the way. The sample points then include the position and orientation of the vessel so that it can be constructed at each sample point. The vessel is a polygon with five corners, so the safe area around the vessel at each sample point on the path is also constructed to be a polygon with five corners. See Steen and Zetterqvist (2021) for more details. The corners of the safe area extend out from the vessel and have a maximum length of 500 meters if there is no obstacle in the way. If an obstacle is in the way, the safe area will extend until it touches the obstacle, as long as it keeps the polygon convex. A map from an archipelago, where the area around the islands has been marked as obstacles, is used to test out the functionality of the safe area formulation. Figure 5.1.1 shows the part of the map used in the results and how the safe area is formulated around each sample point of the path.

For each x_k , the position of the vessels corners are retrieved as $p_{k,i}$ for $i = 1, \dots, 5$. To get the convex polygon around the vessel, the two convex polygons corresponding to the points on the path before and after the closest point to the path are retrieved. Five new corners are created by linearly combining the corresponding corners of the two polygons. These five corners create a new convex polygon which can be described by two vectors, A_k and b_k such that a corner p_k is within the polygon if $A_k p_k \leq b_k$. Thus the following constraint

$$A_k p_{k,i} \leq b_k, \quad i = 1, \dots, 5, \quad k = 1, \dots, N. \quad (5.2)$$

is added to the problem formulation (5.1).

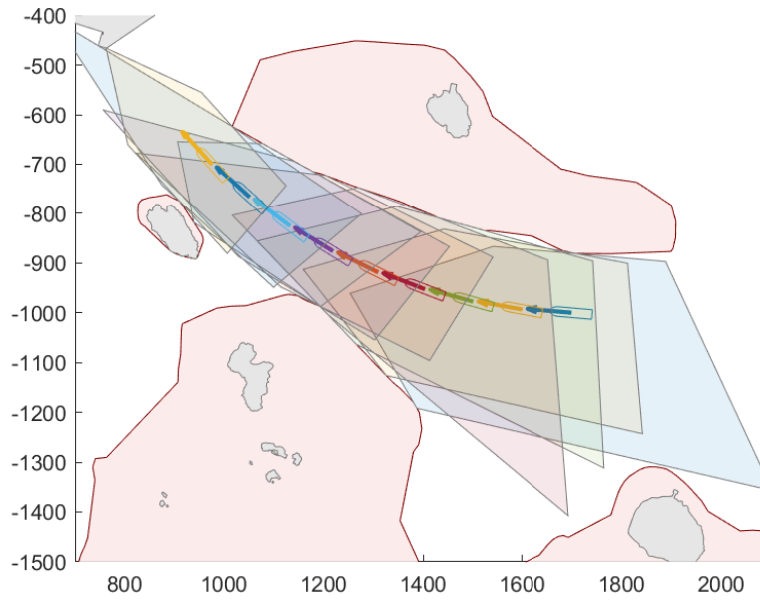


Figure 5.1.1: The path is simulated with a -6 degree angle on both thrusters. The polygons represent the safe area and are restricted by the islands surrounding the path.

5.2 Results

The vessel braking within the safe area, shown in Figure 5.1, is discussed in this section. Figure 5.2.1 shows how the vessel travels within the safe area. It does not follow the path anymore, as there is nothing in the formulation that requires it to be close to the path at all. Instead, it travels mostly in a straight line and stays well within the safe area. Since the path was simulated using a -6° angle, the initial position of the thruster angles will be the same, $\alpha_1 = \alpha_2 = -6^\circ$. The initial velocities are $u = 10$ m/s, $v \approx -0.69$ m/s and $r \approx 0.0093$ rad/s. The initial position is $\eta = \begin{bmatrix} -1000 & 1700 & -85^\circ \end{bmatrix}^T$.

The vessel travels 257.2 meters and comes to a full stop in 59.0 seconds. The distance is very similar to the straight path distance, as the vessel travels mostly in a straight line here as well. The orientation of the vessel is similar to what it is at the start, which shows that given this formulation of the problem, it is not optimal to turn the ship much.

Figures 5.2.2 and 5.2.3 show the states and controls over time. Now the vessel is turned -85° instead of 0° , and starting at -1000 on the x-axis and 1700 on the y-axis, so the position is different from the formulation in Chapter 4. The surge velocity u looks similar, and there is some change in the sway velocity and yaw rate in the beginning but then converges back to 0. The controls look the same as in the straight path case

and are symmetrical. This symmetry makes the vessel travel in a straight line.

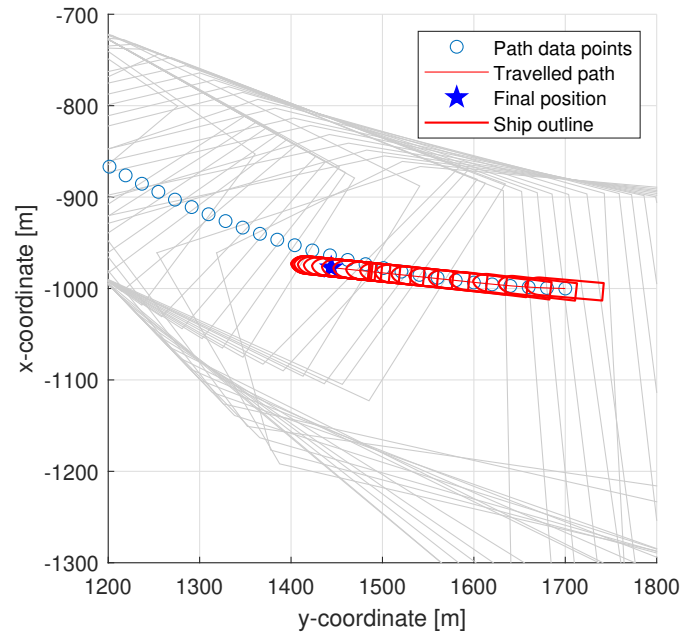


Figure 5.2.1: A simulation of the vessel braking within the safe area. The distance traveled is 257.2 meters, and it takes 59.0 seconds for the ship to come to a complete stop. Since there is no requirement for the vessel to sail along the path but only stay within the safe area, it is able to stop earlier than when following a path.

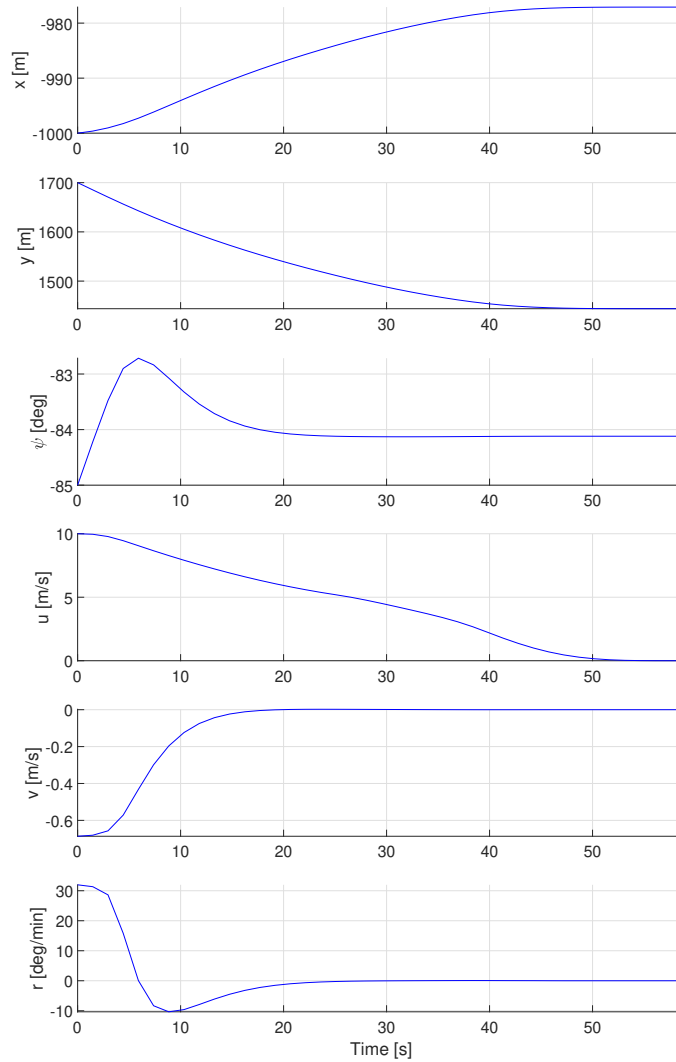


Figure 5.2.2: The plots show the position and velocity of the vessel over time when braking within the safe area. The solution prefers to stop in an as straight line as possible, which can be seen as the solution converges to 0 as fast as possible in sway velocity and yaw rate.

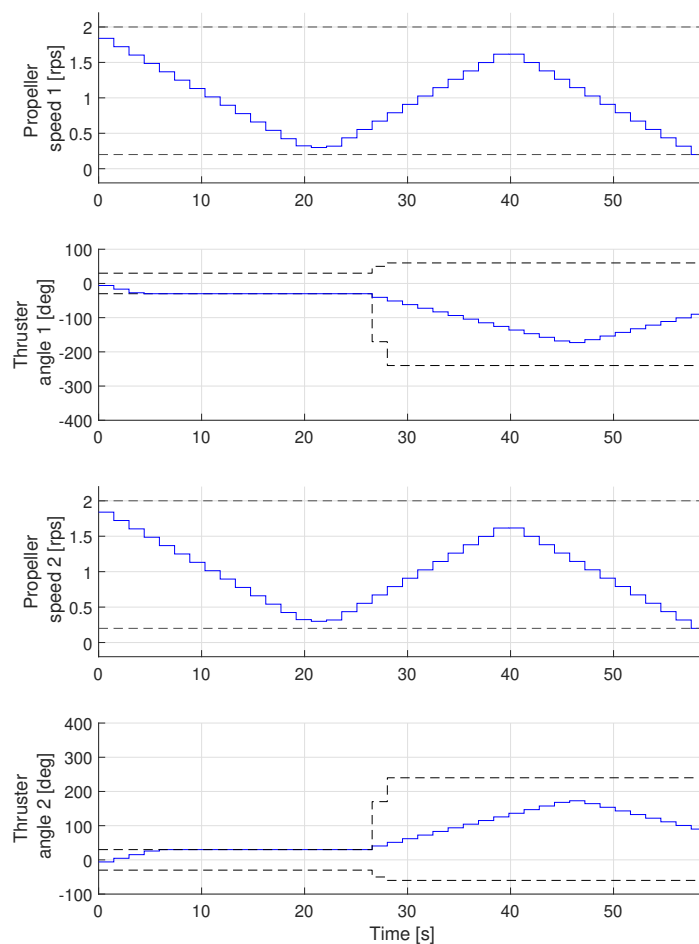


Figure 5.2.3: The plots show control signals of the thrusters, when braking in a safe area. The two thrusters almost act symmetrically to get the result which explain that the vessel travels mostly along a straight line.

Chapter 6

Real-time Formulation

Nonlinear optimal control problems can often be complicated and computationally expensive to solve, and thus solving them can take a long time. Changing the formulation of the problem can often decrease this computational time and even make it real-time. This chapter of the thesis proposes a real-time solution to the optimal braking problem along a path. Two different approaches have been chosen: a receding-horizon approach and a lookup-based approach. The first two sections describe these methods, respectively. The third section compares the two approaches.

6.1 Receding-horizon approach

One approach considered for the real-time formulation of the optimal control problem is a receding horizon approach. As a vessel is sailing along a certain path, at each time point, a crew or an autonomous system would like to be able to know where the vessel would end up if it needed to stop immediately. The optimal braking problem must therefore be calculated frequently to give as much information as possible and be done in real-time for it to be reliable. A receding horizon method is very suitable for this kind of problem. As the vessel is sailing along a path, calculations can be done frequently, and the path ahead will therefore be very similar to the one calculated just a few seconds before. This gives a good base for a very fast solution using the receding horizon algorithm.

The formulation of the optimal control problem is similar to the one defined in Section 4.1, except the cost function changes a bit, with new constants in front of the terms in

order to solve the problem faster. The new cost function is described as

$$\phi(\mathbf{x}_N) + \sum_{k=0}^{N-1} h f_0(kh, \mathbf{x}_k, \mathbf{u}_k) + 100(\hat{\alpha}_1(\mathbf{x}_k) + \hat{\alpha}_2(\mathbf{x}_k)) + 150 \text{dist}_{\text{path}}(\mathbf{x}_k)^2 + 3 \text{dist}_{\text{total}}(\mathbf{x}_k) \quad (6.1)$$

where $\phi(\mathbf{x}_N)$ and $f_0(kh, \mathbf{x}_k, \mathbf{u}_k)$ are discretized from the following continuous functions

$$\phi(\mathbf{x}(t_f)) = 20t_f \quad (6.2a)$$

$$f_0(t, \mathbf{x}(t), \mathbf{u}(t)) = 10(\Delta n_1^2 + \Delta n_2^2) + 10(\Delta \alpha_1^2 + \Delta \alpha_2^2) + 0.01(n_1^2 + n_2^2). \quad (6.2b)$$

A small change in the discretization will be used to speed up the calculations. Instead of using the 4th order Runge-Kutta method for calculating the slope of each discretized interval, a second order Runge-Kutta method, the midpoint method, is used. This halves the number of function evaluations and simplifies calculations for the necessary derivatives used in the optimization problem, but gives similar results. The midpoint method is given by,

$$\mathbf{k}_1 = \mathbf{f}(t_k, \mathbf{x}_k, \mathbf{u}_k) \quad (6.3a)$$

$$\mathbf{k}_2 = \mathbf{f}(t_k + h, \mathbf{x}_k + \frac{h}{2}\mathbf{k}_1, \mathbf{u}_k) \quad (6.3b)$$

$$\mathbf{F}(t_k, \mathbf{x}_k, \mathbf{u}_k) = \mathbf{x}_k + h\mathbf{k}_2. \quad (6.3c)$$

Just like in the 4th order Runge-Kutta method, the midpoint method discretizes the continuous function $\dot{\mathbf{x}} = \mathbf{f}(t_k, \mathbf{x}_k, \mathbf{u}_k)$ such that it becomes $\mathbf{x}_{k+1} = \mathbf{F}(t_k, \mathbf{x}_k, \mathbf{u}_k)$ with the step size h .

Along with decreasing the number of function evaluations, code generation was used to speed up the execution time. The formulation of the optimal control problem was generated into a precompiled header. This compiles the problem into a different form which can then be processed much faster by the computer when running the program. Since the time for executing the function and its derivatives has now decreased significantly, the optimization problem can be run much faster.

After these two steps have been taken to decrease the execution time, the receding horizon approach can minimize the time the solver takes to find an optimal solution. The optimal braking problem is solved for each sample point on the path. After one

problem is solved, the optimal solution is fed into the next problem as an initial guess. The difference in the shape of the path ahead between starting from one sample point to the next is not significant. Therefore, using the optimal solution as an initial guess gives a good warm start to the problem, and the solver does not take long to converge to a new optimal solution. To make sure that the first point does not take too long of a time, it is also given a warm start with an old optimal solution. This can be done by having a record of reasonable solutions given the angle of the path or having a basic straight line solution as the warm start.

Even though the shape of the path does not change a lot from one point to another, the change in position can be significant. The path can therefore be shifted to start the optimization at the origin, with 0 degree heading, no matter where in the n -frame the starting point actually is. The shifting of each point in the path is done with the following equation,

$$\boldsymbol{\eta}_s = \mathbf{R}^T(\psi_0)(\boldsymbol{\eta} - \boldsymbol{\eta}_0) \quad (6.4)$$

where $\boldsymbol{\eta}_s$ denotes the position after it is shifted, $\boldsymbol{\eta}$ denotes the position that is to be shifted, $\boldsymbol{\eta}_0$ denotes the initial position of the vessel, $\mathbf{R}(\psi_0)$ is the rotational matrix, and ψ_0 denotes the angle of the initial position. Using (6.4) for all the points on the path, shifts the whole path so that the initial state always starts at $\boldsymbol{\eta} = (0, 0, 0)$.

For each iteration in the receding horizon approach, the path is shifted before solving the optimization problem. This makes the change in the path even smaller, which in turn allows the solver to solve it even faster. The receding horizon method is defined in Algorithm 1.

Algorithm 1: Receding-horizon method

```

1  $w_0 \leftarrow w_{\text{old}}$ 
2 for  $i = 1$  to number of paths do
3   Current state  $\leftarrow i$ th node of the path
4   Transform the whole path using the current state  $\boldsymbol{\eta}_0$ , and (6.4)
5   Run optimization using the transformed state as initial state, and  $w_0$  as
     initial guess
6   Retrieve  $w$  from the optimization
7    $w_0 \leftarrow w$ 
8   Transform the solution back and return
9 end
```

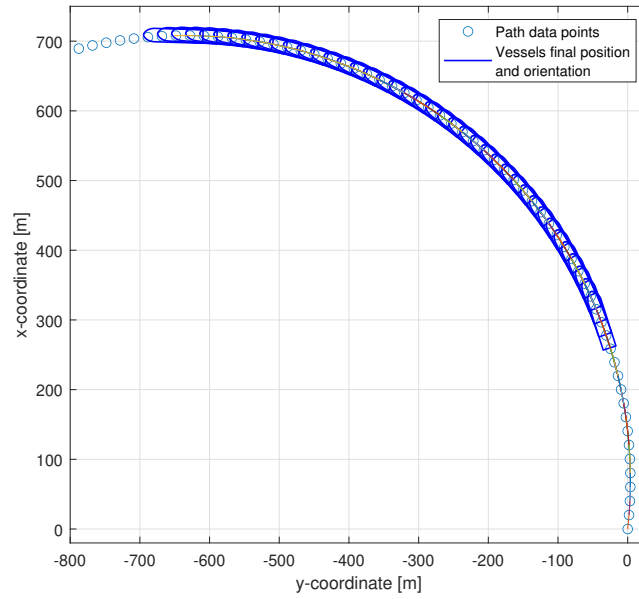


Figure 6.1.1: The vessel's final position and orientation for the receding horizon approach. Each blue outline of a vessel shows the final position and orientation of the vessel after each iteration of the optimization problem. The path was created with a 10 degree angle on both thrusters. This shows that the solution follows the path very well for all the iterations. A total of 40 iterations are shown, starting from the first 40 sample points on the path.

6.1.1 Results

Using the receding horizon method described above, along with decreasing function evaluations and using code generation, the optimal control problem was solved with quite good results, as seen in Figures 6.1.1 and 6.1.2. Figure 6.1.1 shows the final position of the vessel for 40 iterations of the receding horizon method. All the starting points on the path are the same, with the surge speed $u = 10$ m/s and the angle $\alpha_1 = \alpha_2 = 10^\circ$. The corresponding sway velocity $v \approx 1.11$ m/s, yaw velocity $r \approx 0.014$ rad/s, and propeller speed $n_1 = n_2 \approx 1.89$ rps are chosen to keep the vessel at steady state. Since the path was retrieved by simulation, all the solutions in the receding-horizon method should look the same after the transformation. In Figure 6.1.1, it can be seen that the solution is very well behaved and follows the path closely for all iterations.

Figure 6.1.2 shows the computational time for each iteration with three different initial guesses and the corresponding optimal distance. The results show that both the computational time and the distances depend on the initial guess, as the results are different with different initial guesses. Three different initial guesses are used to show

this; the optimal solutions from following a straight path, a curved path simulated with 5 degree angle on each thruster, and a curved path simulated with a 10 degree angle on each thruster.

The 5 degree solution takes the longest time to solve but gives the shortest distance. The computational time for the straight-line solution is in the middle of the other two but gives the longest total stopping distance. Both of these solutions take a rather long time to solve in the first iteration, but this is expected as the problem needs to adjust from the straight or 5 degree curve to the 10 degree curve. The 10 degree solution does not take a longer time in the first iteration, like the others, which is also expected as it is already an optimal solution for the curve used. This solution also gives the best computational time in the other iterations, and the distance is in between the other two solutions.

These results show, as mentioned above, that the solution is dependent on the initial guess. This is not ideal as one would want all the solutions to converge to the same solution which gives the same distance. Even though converging to the exact same solution at all times is ideal, the solution is still quite good as there are only minor differences between the distances and the computational time is almost always under 0.2 seconds. The sample time between points on the path is 2 seconds which makes this solution well within real-time feasibility.

To show that the receding-horizon method also works in more realistic scenarios, different paths were tested. These paths were also simulated using the vessel dynamics but made more realistic by having the vessel sail between islands. This way, each point on the path has different velocities and thruster configurations. The different paths can be seen in Figure 6.1.3, which shows that the solutions from the receding horizon algorithm follow the path closely. Since all the solutions follow the path this closely, none of them hit any obstacles. Figure 6.1.4 shows the computational times and total distances.

Even with different initial values at each point on the path, and slightly different shapes of the path ahead, the problem can be solved in real-time. The sample time is two seconds, and the problem is mostly solved in between 0.1 and 0.4 seconds. The total stopping distances depend both on the initial values and the curvature of the path. Both of the paths lower the velocity of the vessel a little along the way, which explains the decreasing distance with later iterations, but a slightly larger curvature on the path

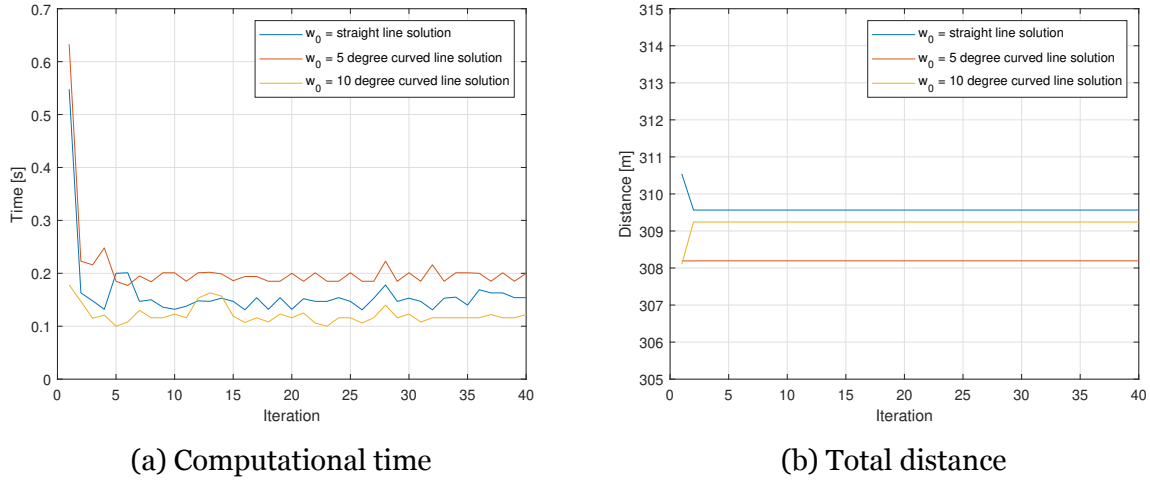


Figure 6.1.2: (a) shows the computational time of the optimization problem for each iteration in the receding horizon method, and (b) shows the optimal distances after each iteration. Three different initial guesses were tested, the blue line shows the time with an initial guess of a straight line solution, the red line shows an initial guess of a curved line solution with a five degree turn, and the yellow line shows an initial guess of a curved line solution with a ten degree turn. The computational times and the total distances depend on the initial guess, but in general the computational times are fast, and the total distances are similar.

points from 20 to 40 explains the sudden increase in the total distance.

6.2 Lookup-based approach

Another approach considered for the real-time formulation of the optimal control problem is a lookup-based approach. The idea is that instead of calculating the optimal control problem at each step, a database of optimal solutions can be built by running the optimal problem multiple times, using different starting velocities and thruster angles. As the vessel travels along a path, an optimal solution can be retrieved from the database. The solution must have similar initial velocities and thruster angles as the actual state of the vessel at each point and also be sufficiently close to the path ahead.

At each data point on the path, the path is transformed using (6.4). The solutions in the database all start from the origin, and thus the path needs to be transformed such that it can be compared to the solutions. All the solutions in the database are then considered and the cost of each path is calculated. The cost consists of the distance from the path, the total distance traveled, and the deviation from the initial state. Once the cost for

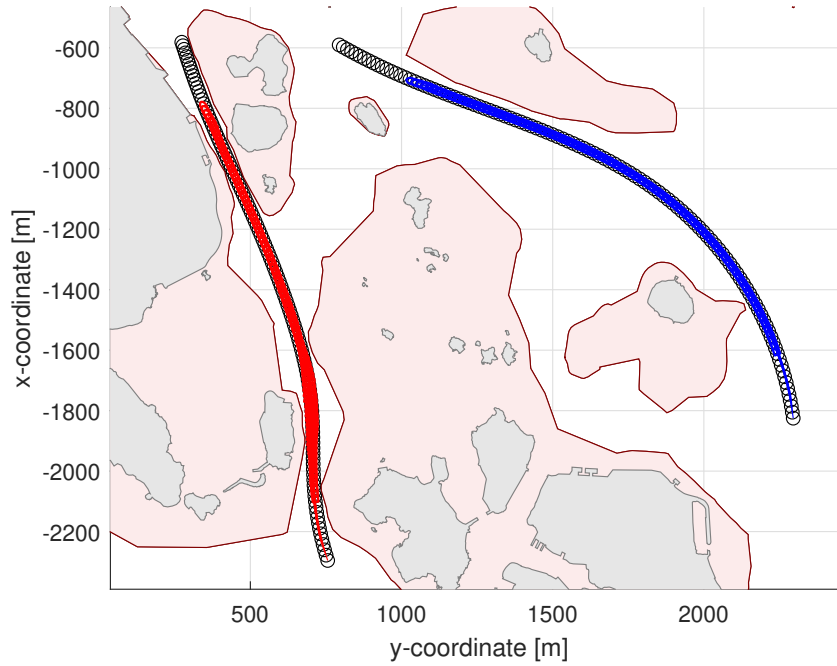


Figure 6.1.3: The receding-horizon algorithm tried on two different, more realistic paths. Each red or blue outline shows the final position of the vessel after each iteration. All of the solutions on both paths follow the path very well, and stops without hitting any obstacles.

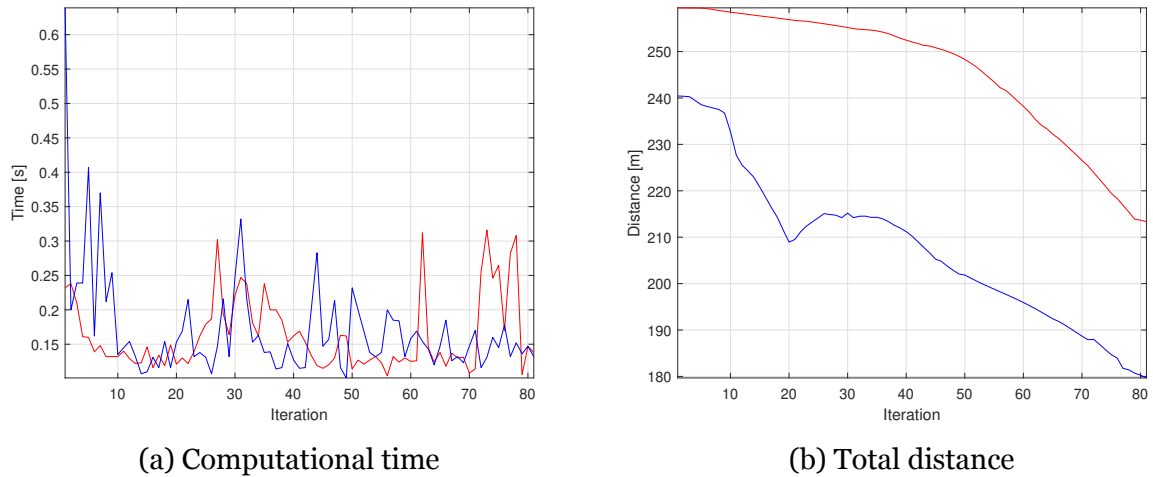


Figure 6.1.4: (a) shows the computational times of each of the paths in figure 6.1.3 the optimization problem for each iteration in the receding horizon method, and (b) shows the optimal distances after each iteration. The blue lines correspond to the path with the blue outline of the ship, and similarly with the red line. The computational times are still well real-time feasible as there are two seconds between each sample point on the path. The total distances depend on the initial values, and in this case, the initial velocities are lowering as the vessel sails within tighter spaces. The curve of the path is also important as it can be seen on the blue path, at iteration 20, where the curve increases slightly, and the distance also increases a bit.

all solutions has been calculated, the solution corresponding to the minimum cost is transformed and returned. These steps are followed for each data point on the path. The lookup-based method is defined in Algorithm 2.

Algorithm 2: Lookup based method

```

1 for  $i = 1$  to number of data points on path do
2   Current state  $\leftarrow i$ th node of the path
3   Transform a part of the path using the current state  $\eta_0$ , and (6.4)
4   for  $j = 1$  to number of solutions in database do
5     Cost = distance to the path + total distance + deviation from the initial
        state
6     Pick the solution with minimum cost
7   end
8   Transform and return the solution
9 end

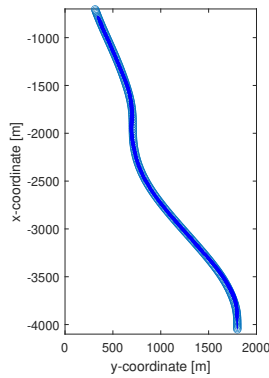
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6.2.1 Results

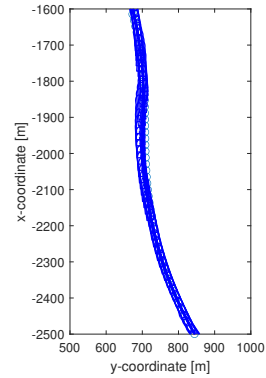
The optimal control problem was solved using the lookup-based approach with decent results. Figure 6.2.1 shows the path the approach was tested on and the final position of the solution after each iteration. Figure 6.2.1a shows the full path and Figure 6.2.1b shows a part of the path. The lookup-based approach works well for some starting data points but not all of them. When the initial value of the data point has low velocity and a large curvature, the approach does not give very good solutions. This can be seen in Figure 6.2.1 (b), where the final point of the vessel is entirely off the path. This means that the vessel is deviating about 20-30 meters off the path. The reason why this is happening is because of the database that was gathered. The database was gathered such that the greater curvatures were when the vessel had high velocity. A different database might be able to handle these cases better.

Figure 6.2.2 shows the computational time and total distance traveled for each iteration. The computational times for the lookup-based method are very good, where almost all iterations were solved within 0.1 and 0.2 seconds. The total distance traveled was a little bit inconsistent and not as smooth as in the receding-horizon approach. There are also some irregular peaks happening for some of the iterations. This is, however, just dependent on the database, where the solver picks the most appropriate solution, and there is no dependence between each iteration.

The database in use contained 620 optimal solutions, which the algorithm had to loop



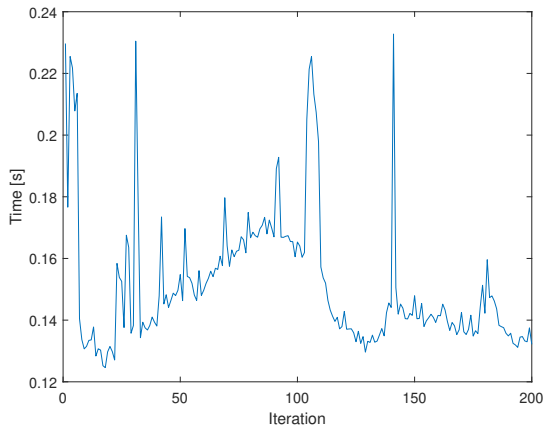
(a) Full path



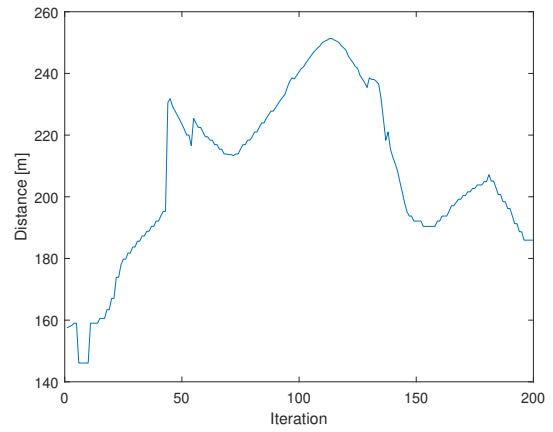
(b) Part of the path

Figure 6.2.1: (a) shows the full path tested and (b) shows the path zoomed in. The lookup-base method works well in most of the cases. But in some cases like shown in (b) the vessel goes off the path completely, which corresponds to going off the path about 20-30 meters.

over. Increasing and decreasing the number of solutions in the database will affect the computational time. Changing the database can also affect how good the solution will be since having a lot of different curvatures and velocities will give the solver more accurate solutions to pick out from.



(a) Computational time



(b) Total distance

Figure 6.2.2: (a) shows the computational times for the path shown in Figure 6.2.1, and (b) shows the total distance of each iteration in the same path. The computational times are quite fast, and in general faster than in the receding horizon method. The change in the total distance between iterations are not as smooth as in the receding horizon method and there are irregular peaks in the distances.

Chapter 7

Conclusions

In this work, two optimal braking problems were formulated and implemented, braking along a path and braking within a safe area. A ship model with 3 degrees of freedom was used to simulate the vessel's movements on water. The controllable forces acting on the vessel were created with two azimuth thrusters, while the environmental forces were neglected for simplicity. The optimal braking problem was discretized using a fourth-order Runge-Kutta method and solved using a direct multiple shooting method. The cost function of the optimal problem aimed to minimize, for example, the final time, the total distance, and when braking along a path, the distance to the path. The cost function was constrained subject to the dynamics and a set of constraints on both the states and the controls. Upper and lower bounds were put on the propeller speed and the thruster angle, where the constraints on the thruster angle depended on the total velocity of the vessel. The result showed that both optimal control problems gave good results. The simulation showed that the vessel stayed close to the path in the optimal braking problem along a path and stayed within the safe area in the optimal braking problem within a safe area. The results also showed that when the vessel was restricted to staying close to the path, it traveled further than in the safe area problem, as well as with greater curvature of the path, the vessel traveled further.

Once both problems were formulated, the braking along a path problem was further used to create a real-time formulation. Two approaches were tested, a receding-horizon approach and a lookup-based approach. Both approaches were solved with real-time feasible computational times. The sample time was two seconds, and the computational times for both approaches were mostly between 0.1 and 0.4 seconds,

which shows that the computational times were good compared to the sample time. The results from the receding-horizon approach showed that the solution in each iteration was always close to the path and had a total distance similar to the iteration before. The computational time for each iteration was higher than in the lookup-based approach and a bit more irregular as it differed quite a bit between each iteration. One drawback of the receding-horizon approach is that the solution depends on the initial guess. However, the solutions were still close enough to be considered a good approach. The results from the lookup-based approach showed that the computational times were, in general, faster than in the receding-horizon approach. For the same path, using the same database, the approach would also always give the same solution. The solution depends on the database, as the database used in this work was gathered from turns at high velocities. Testing the approach on different paths showed that large turns of the path at low velocities did not give good solutions while sailing at higher velocities did. The total distance from one iteration to the next was not as smooth with this approach, which was to be expected as the two iterations do not correlate at all.

The methods studied here are crucial for optimal braking of a vessel in emergency situations. The results of this work are an important contribution to future safe shipping.

7.1 Future work

In this thesis, optimal braking problems and real-time solutions were formulated. Two different real-time approaches were tested, and both approaches resulted in a good, real-time feasible solution. However, both approaches had some limitations that would be worth looking into to improve the solutions. The receding-horizon approach was dependent on the initial guess in the first iteration, and the lookup-based approach was dependent on the database used. The database used for this thesis showed to be insufficient for turns with a short radius at low velocities. Improving these limitations would strengthen both the real-time solutions. Additionally, even though the computational times are good and real-time feasible, some changes in the implementation could potentially lead to faster computational times.

Out of the two optimal braking problems studied, the optimal braking on a path and optimal braking within a safe area, only the first one was studied in regards to the real-

time solution. What is yet to do is to formulate a real-time approach for the safe area solution. This could be an interesting problem because, as we could see in the results, the safe area solution can brake within a shorter distance than the path solution. The goal is to brake safely in the shortest distance possible, so the solution could become even better.

This work focuses entirely on optimal braking with two azimuth thrusters. Solving the problem using different thruster configurations to understand how the problem scales in complexity and computational time would be interesting to study.

As collision avoidance is defined as doing whatever is necessary to avoid a collision, braking close to a path is only one of the possible approaches in each situation. The different approaches must be evaluated together to determine the best action to take. Therefore, merging the approaches described in this thesis with other collision avoidance tools and picking the best approach for each situation would be interesting for future work.

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